Air Power:
Ground Unit and Ground Target Data

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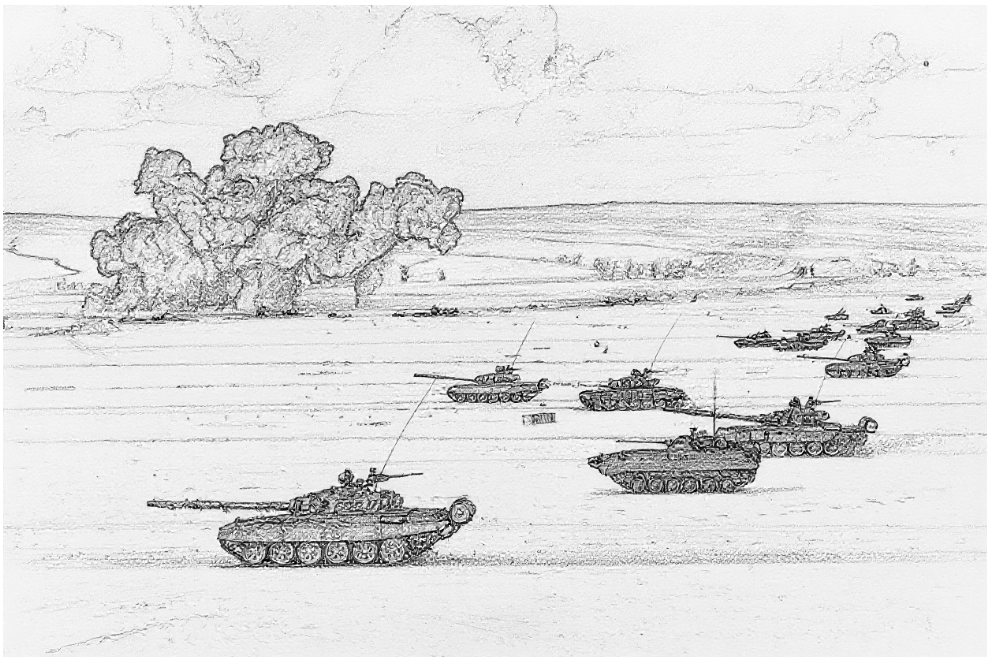


INTRODUCTION

For many, the principal focus of the *Air Power* game is air-to-air combat. That said, no less an authority on air-to-air combat than Robin Olds stated:

> "You're not going to win a war by shooting down a damn MiG, you're going to win a war by bombing the bejesus out of whatever you were sent to bomb."[1]

Here we deal with those "whatevers," the ground units and ground targets that are the targets of air-to-ground attacks, and some of which shoot back.

This document first presents ground combat and transport units, then air-defense units (AAA, FCR, SAM, EWR, and CCU units), and finally ground targets. An appendix gives some examples of battalion-sized units.

Much of the data here is adapted from *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*, from *Air Power Journal*, and from Malcolm Pipe's extensive compendium of AAA and SAM units. However, the data have been carefully revised to improve their accuracy and internal consistency, a wider variety of ground combat units are covered to give a richer experience, and the descriptions of the units and their weapon systems have been expanded.

Extensive notes accompany the main text, providing provenance of data, explanations of changes, sources of information, and justifications for decisions.

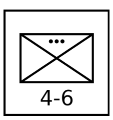
GROUND COMBAT AND TRANSPORT UNITS

Ground combat units are formations of infantry, tanks, IFVs, APCs, other AFVs, or artillery whose primary role is combat with other ground units. Transport units provide strategic mobility to infantry and towed artillery and are the life-blood of logistic networks.[2]

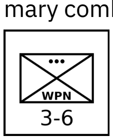
Table 1 summarizes the properties of ground combat and transport units, and Figure 1 shows their representation as counters. All platoons and batteries in Table 1 are also available as sections and squads. AFVs have heavy, medium, light, or open armor protection. The AAA class is B2 for infantry platoons and B3 for AFV platoons, indicating that they are capable of barrage fire to altitudes of 2 and 3 levels, respectively. Infantry and AFV sections and squads are not capable of barrage fire.

Table 2 gives an extensive but not exhaustive list of vehicle types. It should serve as guidance for other vehicles.

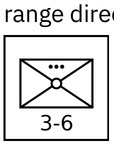
Infantry Units: Infantry are soldiers trained to fight on foot. In some cases, they are also trained to fight from IFVs. Their primary roles in modern warfare are seizing, clearing, and holding ground, often as part of a combined-arms formation with tanks and artillery, highly mobile offensive and defensive operations in combination with helicopters, and combat in difficult terrain, including urban areas, mountainous zones, and dense jungle or forest. An important secondary characteristic of infantry is any associated transport. Motorized infantry units are transported by trucks, giving them operational mobility. Mechanized infantry are transported by APCs or IFVs, giving them both tactical and operational mobility. Light infantry units have no or fewer supporting vehicles, although modern light infantry is often airborne.



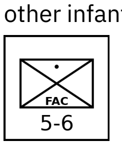
Infantry Platoon: Infantry platoons are armed principally with small arms, often supplemented with light machine guns and sometimes with light grenade launchers and portable anti-tank weapons such as bazookas, LAWs, MAWs, and portable ATGMs. They are the primary combat units in larger infantry formations.



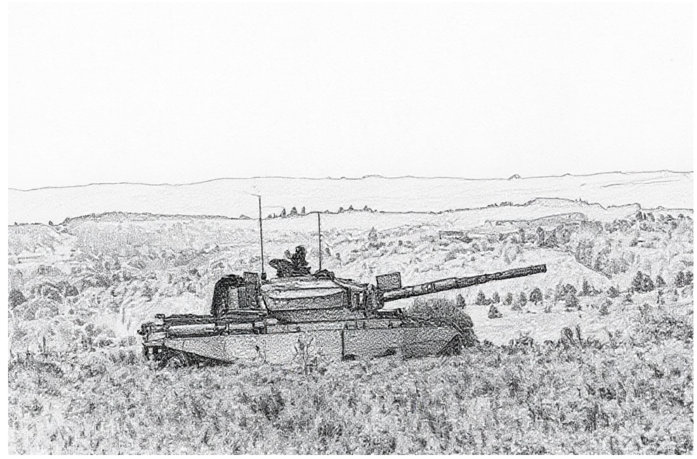
Infantry Weapons Platoon: These are equipped with a mix of heavier weapons, such as medium or heavy machine guns, recoilless rifles, portable ATGMs, and automatic grenade launchers. They are often battalion-level assets, and their role is to provide longer-range direct-fire support to the associated infantry units. [3]

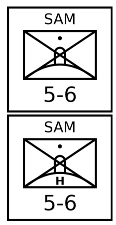


Infantry Mortar Platoon: These are equipped with somewhat portable 60-mm, 81-mm, 82-mm, or similar mortars. They are often battalion-level assets, and their role is to provide indirect fire support and smokescreens for associated infantry units. Due to the weight of ammunition required for sustained fire, an infantry mortar platoon will often be transported by a light truck platoon even if the other infantry units in its formation do not have transport.



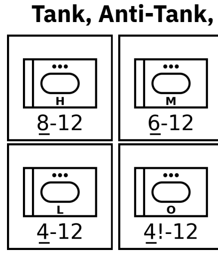
Infantry FAQ Squad: An infantry FAC squad has a specialized forward air controller with other soldiers providing communication and protection. An infantry FAC squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



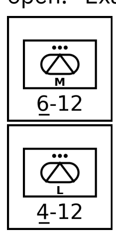


Infantry SAM and Infantry Heavy SAM Squads: Infantry SAM and infantry heavy SAM squads are equipped with shoulder-launched and pedestal-launched SAMs, respectively, with one launcher per squad. Their SAM capabilities and VP values are given in a subsequent section. An infantry SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.

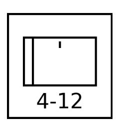
Tank, Anti-Tank, and Reconnaissance Units:



Tank Platoon: Tanks are armored vehicles that are specialized for direct fire with a gun. Here, the gun does not have to be mounted in a turret; tank destroyers with casemate guns such as the SU-100, Kanonenjagdpanzer, and Stridsvagn 103 are also considered to be tanks. Tanks may be heavy,[4] medium,[5] light,[6], or open.[7] Examples are given in Table 2.



Anti-Tank Missile Platoon: Anti-tank missile vehicles are armored vehicles equipped with anti-tank guided missiles which mission is the destruction of other armored vehicles. Conventional gun-armed tank destroyers are considered to be tanks.[8] Anti-tank vehicles may be medium[9] or light.[10] It is often a battalion or regimental asset. Examples are given in Table 2.



Towed Anti-Tank Gun Battery: These have anti-tank guns in towed mounts with only very limited protection. A truck, tracked carrier, or APC platoon may transport a towed anti-tank battery. It is often a regimental or divisional asset.

Reconnaissance Platoon: Reconnaissance vehicles are specialized in scouting and observation and with no significant offensive weapons. They are armed only with machine guns. Vehicles used for reconnaissance that do have significant offensive weapons (e.g, 20 mm or larger guns) are typically classified as light tanks. Reconnaissance vehicles may be light[11] or open.[12] Examples are given in Table 2.

IFV and APC Units: Both IFVs and APCs are armored vehicles that transport infantry. However, IFVs have a 20 mm or larger cannon for fire support, and APCs do not. Some IFVs are also designed to allow their attached infantry to fight while mounted.[13] A mechanized infantry platoon will normally be transported by a platoon of APCs or IFVs.

IFV Platoon: IFVs can be medium[14] or light.[15] Examples are given in Table 2.

APC Platoon: APCs can be heavy[16]. medium[17] light,[18] or open.[19] Examples are given in Table 2.

Armored-Car and Half-Track Units: Armored cars and armored half-tracks are considered to be light tanks, light anti-tank vehicles, light reconnaissance vehicles, light IFVs, or light or open APCs according to their role and armament. That is, the means of traction — tracked, half-tracked, or wheeled — is a secondary consideration. For example, the AMX-10 RC is considered to be a light tank, the BRDM-2 ATGM variants are considered to be light anti-tank vehicles, the Ferret is considered to be a light reconnaissance vehicle, the LAV-25 is considered to be a light IFV, and the BTR-60 is considered to be a light APC.

HQ Units: These are the HQ platoons of battalions or regiments. Normal platoons serve as the HQ platoons of companies.[20]

Infantry HQ Platoon: An infantry HQ platoon is the HQ unit of an infantry battalion or regiment. The HQ platoon of a mechanized infantry battalion or regiment will consist of an infantry HQ platoon transported by a normal IFV or APC platoon. The HQ platoon of a motorized infantry battalion or regiment will consist of an infantry HQ platoon transported by a normal truck platoon.

Mobile HQ Platoon: A mobile HQ platoon is typically the HQ unit of a battalion or regiment other than those of infantry, tank, mechanized infantry, or armored artillery. It is equipped with trucks, and its personnel are not specialized in fighting as infantry.

Armored HQ Platoon: An armored HQ platoon is the HQ unit of a tank, mechanized infantry, or armored artillery battalion or regiment. It is typically equipped with modified APCs or IFVs.

Artillery Units: Artillery is specialized in indirect fire. The weapons may be mortars, guns or howitzers, rockets, or missiles and the mounts may be armored, tracked, mobile, or towed.

Armored Mortar Platoon: These have mortars mounted in armored vehicles that provide both mobility and protection for their crew and weapons from small arms and artillery shrapnel. Examples are given in Table 2.[21]

Towed Mortar Platoon: These have 120 to 160 mm mortars on unprotected, towed mounts. Smaller infantry mortars are found in infantry mortar platoons. A towed mortar battery will normally be transported by a truck or APC platoon.

Armored Artillery Battery: These have guns or howitzers in armored vehicles that provide both mobility and protection for their crew and weapons from small arms and artillery shrapnel. Armored artillery vehicles have light[22] or open[23] armor. Examples are given in Table 2.

Tracked Artillery Battery: These have guns or howitzers mounted on tracked vehicles that provide mobility, but do not give at most limited protection from small arms or artillery shrapnel.[24] Examples are given in Table 2.

Mobile Artillery Battery: These have howitzers mounted on wheeled vehicles that provide mobility, but give at most limited protection from small arms or artillery shrapnel.[25] Examples are given in Table 2.

Towed Artillery Battery: These have guns or howitzers in unprotected, towed mounts. A truck, tracked carrier, or APC platoon may transport a towed artillery battery.

Armored Rocket Artillery Battery: These are rocket-armed equivalents of armored artillery batteries. Armored rocket artillery vehicles have light[26] armor. Examples are given in Table 2.

Mobile Rocket Artillery Battery: These are rocket-armed equivalents of mobile artillery batteries.[27] Examples are given in Table 2.

Tracked Missile Artillery Battery: These are missile-armed equivalents of tracked artillery batteries.[28] Examples are given in Table 2.

Mobile Missile Artillery Battery: These are missile-armed equivalents of mobile artillery batteries. Examples are given in Table 2.[29]

Transport Units: Trucks and tracked carriers are unarmored or incompletely armored vehicles that transport infantry, towed weapons, and supplies, including fuel and ammunition. Trucks are wheeled and have good or poor mobility, whereas tracked carriers have unrestricted mobility.

Truck Platoons: These are platoons with medium and large trucks. A motorized infantry unit will normally be transported by a similarly-sized unit of trucks. A towed artillery battery will normally be transported by a truck platoon.

Light Truck Platoons: These are platoons with light trucks such as the Jeep, GAZ-69, Humvee, Land Rover, or UAZ-469.

Tracked Carrier Platoons: These are platoons with tracked carrier vehicles from Table 2.[30]

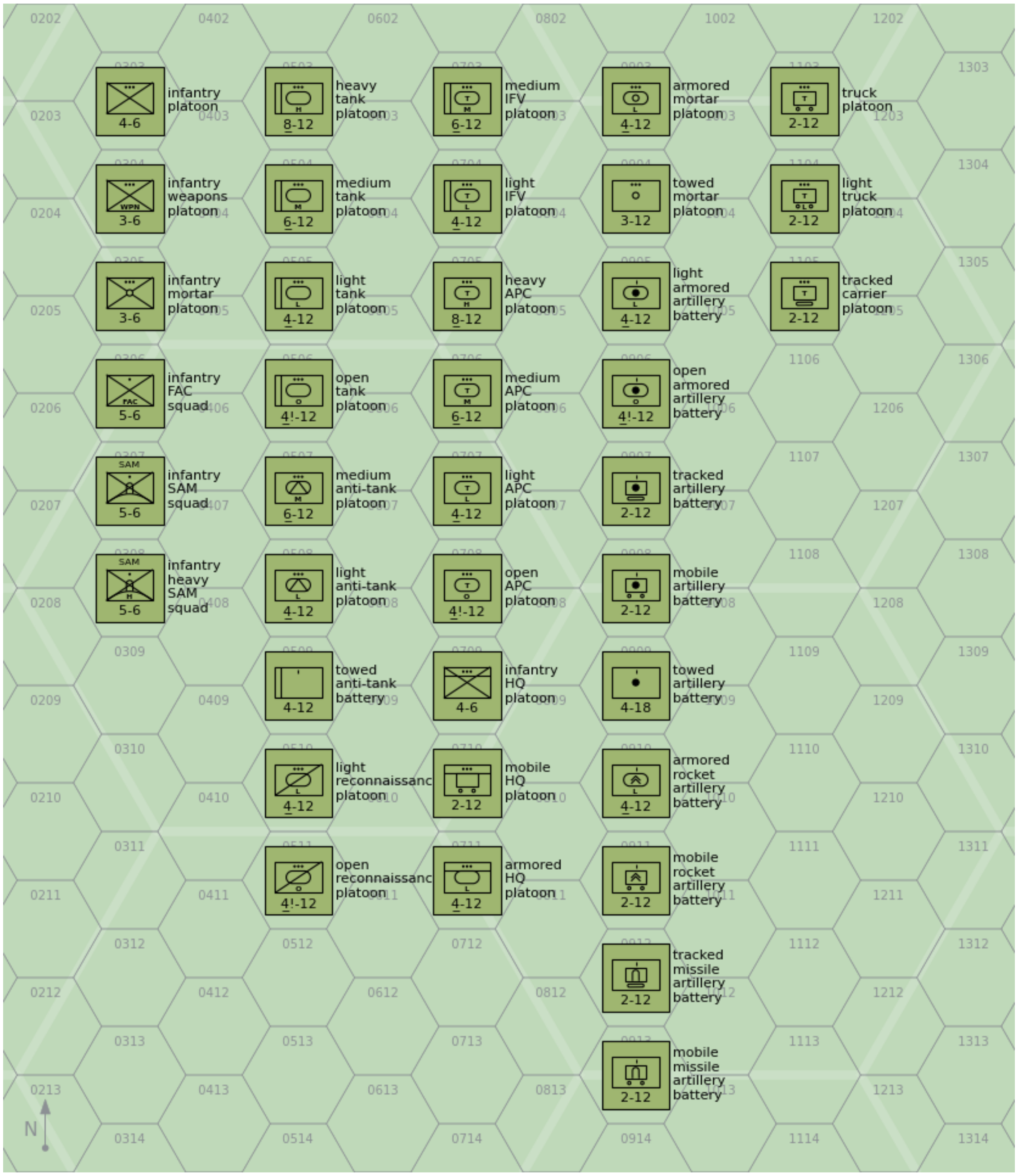


Figure 1: Ground Combat and Transport Units

Table 1: Ground Combat and Transport Units

Type	Size[a]	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class[b]
Infantry	Platoon	4	6	U	5/3/2	B2
Infantry Weapons	Platoon	3	6	U	6/4/2	B2
Infantry Mortar	Platoon	3	6	U	6/4/2	B2
Infantry FAC	Squad	5	6	U	–/–/8	—
Infantry SAM	Squad	5	6	U		—
Infantry Heavy SAM	Squad	5	6	U		—
Heavy Tank	Platoon	$\underline{8}$	12	U	12/8/4	B3
Medium Tank	Platoon	$\underline{6}$	12	U	10/7/3	B3
Light Tank	Platoon	$\underline{4}$	12	U	6/4/2	B3
Open Tank	Platoon	$\underline{4}$!	12	U	5/3/2	B3
Medium Anti-Tank	Platoon	$\underline{6}$	12	U	12/8/4	B3
Light Anti-Tank	Platoon	$\underline{4}$	12	U	10/7/3	B3
Towed Anti-Tank	Battery	4	12	T	8/5/3	—
Light Reconnaissance	Platoon	$\underline{4}$	12	U	6/4/2	B3
Open Reconnaissance	Platoon	$\underline{4}$!	12	U	5/3/2	B3
Medium IFV	Platoon	$\underline{6}$	12	U	10/7/3	B3
Light IFV	Platoon	$\underline{4}$	12	U	6/4/2	B3
Heavy APC	Platoon	$\underline{8}$	12	U	10/7/3	B3
Medium APC	Platoon	$\underline{6}$	12	U	8/5/3	B3
Light APC	Platoon	$\underline{4}$	12	U	6/4/2	B3
Open APC	Platoon	$\underline{4}$!	12	U	5/3/2	B3
Infantry HQ	Platoon	4	6	U	10/7/3	B2
Mobile HQ	Platoon	2	12	G/P	10/7/3	—
Armored HQ	Platoon	$\underline{4}$	12	U	12/8/4	B3
Armored Mortar	Platoon	$\underline{4}$	12	U	10/7/3	B3
Towed Mortar	Platoon	3	12	T	7/5/2	—
Light Armored Artillery	Battery	$\underline{4}$	12	U	12/8/4	B3
Open Armored Artillery	Battery	$\underline{4}$!	12	U	12/8/4	B3
Tracked Artillery	Battery	2	12	U	10/7/3	—
Mobile Artillery	Battery	2	12	G/P	9/6/3	—
Towed Artillery	Battery	4	18	T	8/5/3	—
Armored Rocket Artillery	Battery	$\underline{4}$	12	U	12/8/4	B3
Mobile Rocket Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Missile Artillery	Battery	2	12	U	10/7/3	—
Mobile Missile Artillery	Battery	2	12	G/P	9/6/3	—
Truck	Platoon	2	12	G/P	4/3/1	—
Light Truck	Platoon	2	12	G/P	4/3/1	—
Tracked Carrier	Platoon	2	12	U	4/3/1	—

[a] Platoons and batteries are available as sections and squads with corresponding modifications to their damage resiliences. [b] Platoons with AAA class of B2 and B3 can employ barrage fire to altitudes of 2 and 3 levels, respectively.

Table 2: Vehicle Types

Heavy Tanks

Al-Khalid/VT-1A	PK/CN	2001
Ariete	IT	1995
Arjun	IN	2004
Challenger 1	GB	1983
Challenger 2	GB	1998
Chieftain	GB	1967
Conqueror	GB	1955
IS-2	SU	1944
IS-3	SU	1945
K1	KR	1987
K2 Black Panther	KR	2014
Leclerc	FR	1992
Leopard 2	DE	1979
M-84	YU	1985
M1 Abrams	US	1980
Merkava 1	IL	1979
Merkava 2	IL	1983
Merkava 3	IL	1989
Merkava 4	IL	2004
PT-91	PL	1995
Stridsvagn 103	SE	1967
T-10	SU	1954
T-14 Armata	RU	2024
T-64	SU	1966
T-72	SU	1973
T-80	SU	1976
T-84	UA	1999
T-90	RU	1992
Type 10	JP	2012
Type 90	JP	1990
Type 99	CN	2001

Medium Tanks

AMX-30	FR	1966
ASU-85	SU	1959
Centurion	GB	1945
Comet	GB	1944
Cromwell	GB	1944
ISU-122	SU	1944
ISU-152	SU	1943
Kanonen JPz	DE	1965
Leopard 1	DE	1965
M-50 Sherman	IL/FR	1956
M-51 Sherman	IL/FR	1961
M103	US	1957
M26 Pershing	US	1944
M4 Sherman	US	1942
M46 Patton	US	1949
M47 Patton	US	1951
M48 Patton	US	1952
M60	US	1959
Panzer 61	CH	1965
Panzer 68	CH	1971
SU-100	SU	1944
T-34	SU	1940
T-54	SU	1955
T-55	SU	1958
T-62	SU	1961
TAM	AR	1983
Type 59	CN	1959
Type 61	JP	1961
Type 69	CN	1972
Type 74	JP	1975
Type 79	CN	1984
Type 85	CN	1988
Type 88	CN	1988
Type 96	CN	1997
Type 15	CN	2018
Vickers MBT	GB	1965
Vijayanta	IN/GB	1965

Light Tanks

AMX-10 RC	FR	1981
AMX-13	FR	1953
Fox	GB	1975
Commando (90 mm)	US	1964
LAV I Cougar	CA	1976
LAV II Coyote	CA	1990
M8/Greyhound	US	1943
M24 Chaffee	US	1944
M41 Walker Bulldog	US	1953
M113 C&V with 25 mm	US	1974
M114A2	US	1969
M551 Sheridan	US	1969
Panhard AML	FR	1961
Panhard EBR	FR	1951
PT-76	SU	1951
Saladin	GB	1958
Scimitar	GB	1974
Scorpion	GB	1973
SpPz Luchs	DE	1975
Stryker MGS	US/CA	2006
Strv 74	SE	1958
Type 62	CN	1961
Type 63	CN	1963
Type 05 AAV	CN	2005
Type 08 ARV	CN	2008

Open Tanks

ASU-57	SU	1951
SU-76	SU	1943

Medium Anti-Tank Vehicles

Raketen JPz 1	DE	1961
Raketen JPz 2	DE	1967
Jaguar 1	DE	1978
Jaguar 2	DE	1983

Light Anti-Tank Vehicles

BRDM-2 ATGM	SU	1973
BRDM-3 ATGM	SU	1973
Fennek MRAT	DE/NL	2003
FV438 Swingfire	GB	1971
M150	US	1973
M901 ITV	US	1979
Striker	GB	1976
Stryker ATGM	US/CA	2003
VAB HOT	FR	1978
YP-408 PRAT	NL	1976
YPR-765 PRAT	NL	1980

Light Reconnaissance Vehicles

BRDM-2	SU	1973
Fennek	DE/NL	2003
Ferret	GB	1952
M113 C&V/Lynx	US	1966
M114	US	1962

Open Reconnaissance Vehicles

M20	US	1943

Medium IFVs

BMP-3	SU	1987
Boxer IFV	NL/DE	2011
Strf 9040/CV90	SE	1994
LAV III (add-on-armor)	CA	2009
M2A2/3/4 Bradley	US	1988
M3A2/3/4 Bradley	US	1988
SPz Marder	DE	1971
SPz Puma	DE	2015

Light IFVs

AIFV	US	1979
AMX-10P	FR	1973
BMP-1	SU	1966
BMP-2	SU	1980
Commando (20 mm)	US	1963
FV432 RARDEN	GB	1975
LAV III	CA	1999
LAV-25	CA	1983
M2 and M2A1 Bradley	US	1981
M3 and M3A1 Bradley	US	1981
M113A1 FSC	AU	1972
Piranha IFV	CH	1974
Pbv 301	SE	1961
Pbv 302	SE	1966
SPz Kurz	FR	1959
SPz Lang HS.30	CH	1960
Stryker ICVD	US	2018
Type 86	CN	1992
Type 92	CN	1992
Type 05 IFV	CN	2005
Type 08 IFV	CN	2008
VCBI	FR	2008
Warrior	GB	1987
YPR-765	NL	1977

Heavy APCs

Achzarit	IL	1988
Nakpadon	IL	1998
Namer	IL	2008
Puma	IL	1993

Medium APCs

Boxer APC	NL/DE	2011
Nagmachon	IL	2000
Nagmashot	IL	1990

Light APCs

Bronco/Warthog	SG	2001
BvS10/Viking	SE	2005
BTR-50PK	SU	1958
BTR-60PA/PAI/PB	SU	1965
BTR-70	SU	1972
BTR-80	SU	1986
BTR-152K	SU	1959
Commando	US	1963
FV432	GB	1963
LAV I Grizzly	CA	1976
LAV II Bison	CA	1990
LVTP-5	US	1956
LVTP-7 (AAV)	US	1972
M59	US	1954
M75	US	1952
M113	US	1960
OT-64 SKOT	CS/PL	1963
OT-810 half-track	CS	1958
Piranha APC	CH	1974
Saracen	GB	1952
Saxon	GB	1983
Spartan	GB	1978
Stryker ICV	US/CA	2002
TPz Fuchs	DE	1979
Type 63	CN	1978
Type 77	CN	1978
Type 89	CN	1978
Type 08 APC	CN	2008
VAB	FR	1979
YP-408	NL	1964

Open APCs

AT-P	SU	1954
BTR-40	SU	1950
BTR-152A	SU	1950
BTR-50P	SU	1952
BTR-60P	SU	1959
M3 half-track	US	1941
SKP/VKP	SE	1944
SKPF/VKPF	SE	1956

Tracked Carrier

AT-T/S/L	SU	1950
Bv 202	SE	1964
Bv 206	SE	1980
BvS10/Beowulf	SE	2024
M548	US	1960
Sisu Nasu	FI	1986

Light Armored Mortar Carriers

FV432 with 81 mm	GB	1963
M106	US	1964
M125	US	1964
M1064	US	1990
Stryker MCV	US/CA	2005

Light Armored Artillery

2S1 Gvozdika	SU	1972
2S3 Akatsiya	SU	1971
2S19 Msta-S	SU	1989
AMX-30 AuF1	FR	1977
AS-90	GB	1992
Bkan 1	SE	1967
FV433 Abbot	GB	1965
L-33 Ro'Em	IL	1973
M52	US	1955
M108	US	1962
M109	US	1963
Panzerhaubitze 2000	DE	1998

Open Armored Artillery

AMX-105A	FR	1958
M-50	IL	1962
M7 Priest	US	1942
M40	US	1945
M44	US	1954

Tracked Artillery

2S4 Tyulpan	SU	1975
2S7 Pion	SU	1976
M41	US	1942
M107	US	1962
M110	US	1963
AMX-13 F3	FR	1962

Mobile Artillery

Archer	SE	2016
CAESAR	FR	2004

Light Armored Rocket Artillery

M270 MLRS	US	1983
BM-1	RU	1988

Mobile Rocket Artillery

BM-8	SU	1941
BM-13	SU	1941
BM-31	SU	1944
BM-24	SU	1947
BM-14	SU	1952
BM-21 Grad	SU	1963
BM-27 Uragan	SU	1975
BM-30 Smerch	SU	1989
M142 HIMARS	US	2005

Tracked Missile Artillery

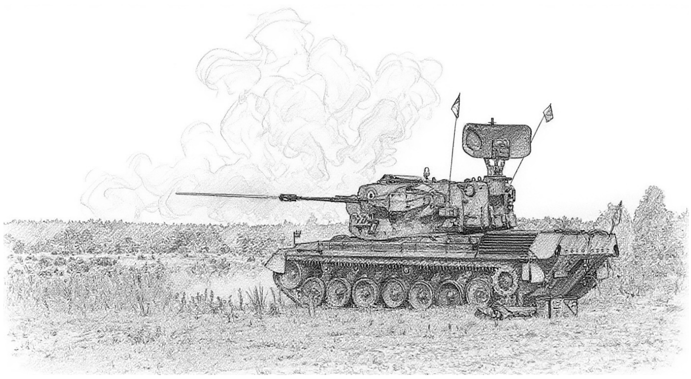
M752 Lance	US	1972
AMX-30 Pluton	FR	1974

Mobile Missile Artillery

9K52 Luna-M (Frog-7)	SU	1962
Corporal	US	1954
Sergeant	US	1962
M34/M50 Honest John	US	1953
M74 Redstone	US	1958
M790 Pershing 1a	US	1962
M1003 Pershing II	US	1983
OTR-21 Tochka (SS-21)	SU	1975
RSD-10 Pioneer (SS-20)	SU	1976

AAA UNITS

AAA units specialize in the use of guns to destroy attacking aircraft, but also have a secondary ground combat capability.



Properties: Table 3 summarizes the properties of AAA units, and Figure 2 shows their representation as counters. All platoons and batteries in Table 3 are also available as sections (but not squads). As well as the properties common to all ground units, Table 3 gives:
[31]

– the AAA class (light, medium, or heavy);
– the maximum altitude;
– the limits of short, medium, and long range in hexes;
– the hit rolls at short, medium, and long range;
– the damage rating;
– whether the unit has an all-weather or ranging-only fire-control radar and its frequency band;
– whether it has night infrared sights; and
– whether it is also equipped with a SAM system.

If the unit has a SAM, this will be described in detail in the section on SAM launcher units.

When comparing these properties, remember that towed batteries have six to eight gun mounts, and armored or mobile sections have only two gun mounts.

FCRs: Some AAA units have integrated FCRs that give them an all-weather and night capability. Most medium and heavy can also use add-on FCRs, described below. Units without an FCR (or with a range-only FCR) are limited to visually sighted targets.

Light AAA Units:

M2 Platoon: The Browning M2 .50 cal (12.7 mm) heavy machine gun is normally found on a low tripod for ground combat, but here is used on a tall dual-purpose mount suitable for air defense. A platoon has six guns.[32]

The M2 gun was introduced in 1933 and has been used by United States forces and their allies in numerous conflicts since then.

DShK-38 Platoon: The DShK-38 12.7 mm heavy machine gun is most commonly found on a low, wheeled mount for ground combat, but here is used on a tall tripod suitable for air defense. A platoon has six guns.[33]

The DShK-38 entered service with the Soviet Army in 1938 and saw action in World War II and numerous conflicts since then, including the Korean and Vietnam Wars, in which the Soviet Union participated or provided arms.

M16 Section: The M16 Multiple Gun Motor Carriage has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a modified M3 half-track. A section has two vehicles.[34]

The M16 was used by United States forces during World War II and (primarily as an anti-personnel weapon) in the Korean War for airfield defense. Additionally, Israeli forces employed it during the 1956 Arab-Israeli War.

M55 Platoon: The M55 has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a towed carriage. A platoon has four gun mounts.[35]

The M55 was used by United States forces during the Korean War and by Pakistani forces during the 1965 India-Pakistan War.

ZPU-1/2/4 Battery: The ZPU-1/2/4 have single, twin, or quad 14.5 mm heavy machine guns in open mounts on towed carriages. A battery has six gun mounts.[36][37][38]

The ZPU-1/2/4 was introduced in 1949 and saw extensive service with Soviet or Soviet-supported forces in many subsequent conflicts, including with communist forces in the Korean War and Vietnam War and with Arab forces in the various Arab-Israeli Wars from 1956 onwards.

Mobile ZPU-1/2/4 Section: This is a section of two light trucks mounting ZPU-1/2/4 guns.[39][40]

Many users of the ZPU-1/2/4 mount it on a suitable truck for increased mobility. Like the towed versions, it has seen combat in many conflicts.

M167 Platoon: The M167 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in an open mount on a towed carriage. The fire-control system has optical sights and a range-only radar. A platoon has four gun mounts.[41]

The M167 was introduced into service with the US Army in 1967, principally for airfield defense, and subsequently served with other US allies.

M163 Section: The M163 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in a lightly armored but open-topped turret on an adapted M113 APC. The fire-control system has optical sights and a range-only radar. A section has two vehicles.[42]

The M163 VADS was introduced into service with the US Army in 1968 and has since been adopted by other US allies. The M163 was used by US forces in the Vietnam War, the 1989 Invasion of Panama, and the 1991 Gulf War. It was also used by the Israel Army in the 1982 Lebanon War and in 2002 during the Second Intifada.

From 1988 in the US Army, each M163 was typically issued with a FIM-92 Stinger launcher with two rounds, so each M163 section effectively carried a mounted FIM-92 section.

TCM-20 Platoon: The TCM-20 is an Israeli adaptation of the M55, with two Hispano-Suiza 20mm guns from obsolete aircraft replacing the four .50 cal machine guns. As with the M55, the fire-control system is entirely optical. A platoon has four gun mounts.[43]

The TCM-20 was employed by Israeli forces from 1970 and subsequently saw combat in the 1973 Arab-Israeli War and the 1982 Lebanon War.

M3 TCM-20 Section: The M3 TCM-20 has a TCM-20 turret mounted in an adapted M3 half-track, like the earlier M16. A section has two vehicles.[44]

The M3 TCM-20 was used by Israeli forces in the 1973 Arab-Israeli War and the 1982 Lebanon War.

Rh-202 Battery: The Rheinmetall Rh-202 has a single or twin 20mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. The most common version is the twin mount. A battery has six gun mounts.[45]

The twin version entered service with West German forces in 1972 and was used for airfield defense. Subsequently, it was exported and notably used by the Argentine Air Force in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).

Panhard M3 DCA Section: The Panhard M3 DCA vehicles have an armored turret for twin 20 mm guns mounted on the Panhard M3 APC. The guns have optical sights, but typically one vehicle in each section or platoon is equipped with a search-and-ranging radar and can share this information automatically with the others. A section has two vehicles.[46]

The Panhard M3 DCA served with the armed forces of Côte d'Ivoire, Niger, and the UAE.

ZU-23 Battery: The ZU-23 has twin 23 mm cannons in an open mount on a towed carriage. It was designed as a more powerful replacement for the ZPU-1/2/4. The fire-control system is entirely optical. A battery has six gun mounts.[47]

The ZU-23 was introduced into service by the Soviet Army in 1960. It later served with the armed forces of many Soviet allies, including with North Vietnam in the Vietnam War, with Egypt and Syria in the 1967 and 1973 Wars, with Soviet forces in the Soviet-Afghan War, with Iraq in the Iraq-Iran War, with Syria in the 1982 Lebanon War, and with both sides in the Afghan Civil War.

Mobile ZU-23 Section: This is a section of two light trucks mounting ZU-23 guns.[48]

As with the earlier ZPU-1/2/4, many users of the ZU-23 mount it on vehicles, including trucks, tracked carriers, and APCs, for better mobility.

Sinai 23 Section: The Sinai 23 has a Dassault TA20 turret with two 23 mm guns and six Ayn-al-Saqr (similar to the Strela-2M) or Stinger missiles on an M113 chassis. It has night IR sights. A section has two vehicles. Each battery operates with one vehicle equipped with an RA20S search radar.[49]

The Sinai 23 entered service with the Egyptian Army in 1991.

ZSU-23-4 Section: The ZSU-23-4 has four radar-controlled 23 mm guns, similar to those in the ZU-23, mounted in an armored turret on a hull adapted from the PT-76 light tank. A section has two vehicles.[50]

The ZSU-23-4 entered service in the Soviet Army in 1965, replacing the ZSU-57-2 and outclassing all NATO anti-aircraft guns at that time. A platoon of four was assigned to the air-defense battery of tank and motor-rifle regiments and later complemented by a platoon of four SA-9 vehicles. It was exported to Soviet allies and saw extensive combat with Arab forces in the 1973 Arab-Israeli War and with Iraqi forces in the Iran-Iraq War and the 1991 Gulf War.

LAV-AD Section: The LAV-AD has a 25 mm GAU-12 rotary cannon and eight FIM-92D Stinger missiles mounted in a turret on the LAV-25 vehicle. It does not have radar, but is equipped with a passive night IR sight. A section has two vehicles.[51]

The LAV-AD entered service with the USMC in 1997, but was retired after only a few years.

M53/59 Section: The M53/59 has a twin M53 30 mm gun mount on a partially armored truck. The fire-control system is entirely optical. A section has two vehicles.[52]

The M53/59 entered service in the Czech Army in 1959 and was used instead of the ZSU-57-2. It was also exported to Iraq and saw combat in the Iran-Iraq War and the 1991 Gulf War.

AMX-30SA Section: The AMX-30SA is equipped with a pair of radar-controlled 30 mm Hispano-Suiza guns mounted in an armored turret on the hull of an AMX-30 tank. It is an improved version of the AMX-30 DCA. A section has two vehicles.[53]

The AMX-30SA served only in the Saudi Army from 1979.

Tunguska Section: The Tunguska has twin radar-controlled 30mm guns and four SA-19 missiles in an armored turret on an armored and tracked hull. It was designed to counter the A-10 and AH-64, against which the earlier ZSU-23-4 had weaknesses. A section has two vehicles.[54]

The Tunguska entered service in the Soviet Army in 1984. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.

Tunguska-M Section: The Tunguska-M is an improved version of the Tunguska with eight missiles instead of four. A section has two vehicles.[55]

The Tunguska-M entered service in the Soviet Army in 1990. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.

Medium AAA Units:

Oerlikon GDF Battery: The Oerlikon GDF has two 35 mm guns in an open mount on a towed carriage. A battery has six gun mounts. The basic fire-control system is optical, but a section is often coupled with the Super Fledermaus or Skyguard fire-control radars. (These radars cannot control a whole battery, only a section.)[56]

The Oerlikon GDF entered service in 1963 and has served widely. Notably, it saw combat in the South Atlantic War with the Argentine forces in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).

Gepard Section: The Gepard has two radar-controlled 35 mm guns in an armored turret on an adapted Leopard 1 tank hull. A section has two vehicles.[57]

The Gepard entered service in 1976 with the West German Army and also served with the Belgian and Dutch armies. After the end of the Cold War, they were retired by these armies, and many were sold on to other countries.

61-K Battery: The 61-K (M1939) has a single 37 mm gun in an open mount on a towed carriage. The fire-control system is entirely optical, and the 61-K cannot be coupled to an external fire-control radar. A battery has six gun mounts.[58]

After entering service in 1939 with the Soviet Army, it served in WW2 and after until replaced by the S-60 57 mm gun. It also served with many Soviet allies, and in particular with communist forces in the Korean and Vietnam Wars.

The 61-K is referred to as the "M-38" in *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*.[59]

Bofors L/60 Battery: The Bofors L/60 has a single 40 mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. A battery has six gun mounts.[60]

It entered service in the 1940s and was widely used by Allied forces during WW2, both on land and sea. After the war, many continued to see service, although it was increasingly inadequate against faster jet aircraft and in many cases was replaced by the new L/70 model.

Bofors L/70 Battery: The Bofors L/70 has a single 40 mm gun in an open mount on a towed carriage. Between the 1930s and the end of WW2, aircraft speed increased substantially, and so the engagement time of the Bofors L/60 became inadequate. The new L/70 fired a lighter shell at a higher velocity and achieved a significant improvement in range and engagement time. The basic fire-control system is optical, but various external fire-control radars can be used with it, including the Super Fledermaus and Flycatcher systems. A battery has six gun mounts.[61]

NATO forces and others adopted it widely starting in 1952.

Bofors L/70 BOFI and BOFI-R Battery: The BOFI version of the Bofors L/70 adds modern sights, a laser rangefinder, and proximity-fuzed shells. The BOFI-R version also adds an integrated fire-control radar. A battery has six gun mounts.[62]

The BOFI is used by the armed forces of Brazil.[63]

S-60 Battery: The S-60 is a Soviet single 57 mm gun on an open mount on a towed carriage. The basic fire-control system is optical, but the gun is often used with the SON-9 (Fire Can) radar. A battery has six gun mounts.[64]

It entered service in 1950 with Soviet forces as a replacement for the 61-K 37 mm gun. It was exported to many Soviet allies and saw combat with Arab forces in the 1967 and 1973 Arab-Israeli War, with communist forces in the Vietnam War, and with Iraqi forces in the Iran-Iraq War and the Gulf War. However, it was not used by communist forces in the Korean War. When paired with the SON-9 (Fire Can) radar, it was considered one of the more dangerous AAA systems faced by US aircraft in Vietnam.

ZSU-57-2 Section: The ZSU-57-2 has two 57 mm cannons in a lightly armored but open-topped turret on an armored and tracked hull. The cannons are similar to the S-60 cannons. The fire-control system is optical. A section has two vehicles.[65]

The ZSU-57-2 entered service in the Soviet Army in 1955 and replaced the BTR-40A and BTR-152A self-propelled 14.5 mm AA guns in the AA batteries of tank regiments. It was in turn replaced by the ZSU-23-4 from 1965. The ZSU-57-2 also served widely with Soviet allies and saw combat with Egypt and Syria in the 1967 and 1973 Wars, with Syria in the 1982 Lebanon War, with North Vietnam in the 1972 Easter Offensive and the 1975 Spring Offensive, and with various factions in the Yugoslav Wars in the 1990s.

Heavy AAA Units:

KS-12 Battery: The KS-12 (M1939 52-K) and the very similar KS-12A (M1944 KS-1) have a single 85 mm gun in an open mount on a towed carriage. The fire-control system is optical, but the guns are often paired with the SON-9 (Fire Can) radar. A battery has six gun mounts.[66]

It entered service with Soviet forces in 1939. After WW2, it began to be replaced in Soviet service by the KS-19 100 mm and KS-30 130 mm guns, but was widely used by Soviet allies and saw combat with communist forces in the Korean War and the Vietnam War.

KS-19 Battery: The KS-19 has a single 100 mm gun in an open mount on a towed carriage. The fire-control system is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. A battery has six gun mounts.[67]

It entered service in 1948 with Soviet forces as a replacement for the KS-12 85 mm gun. It also served with many Soviet allies and saw combat with communist forces in the Korean War and Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.

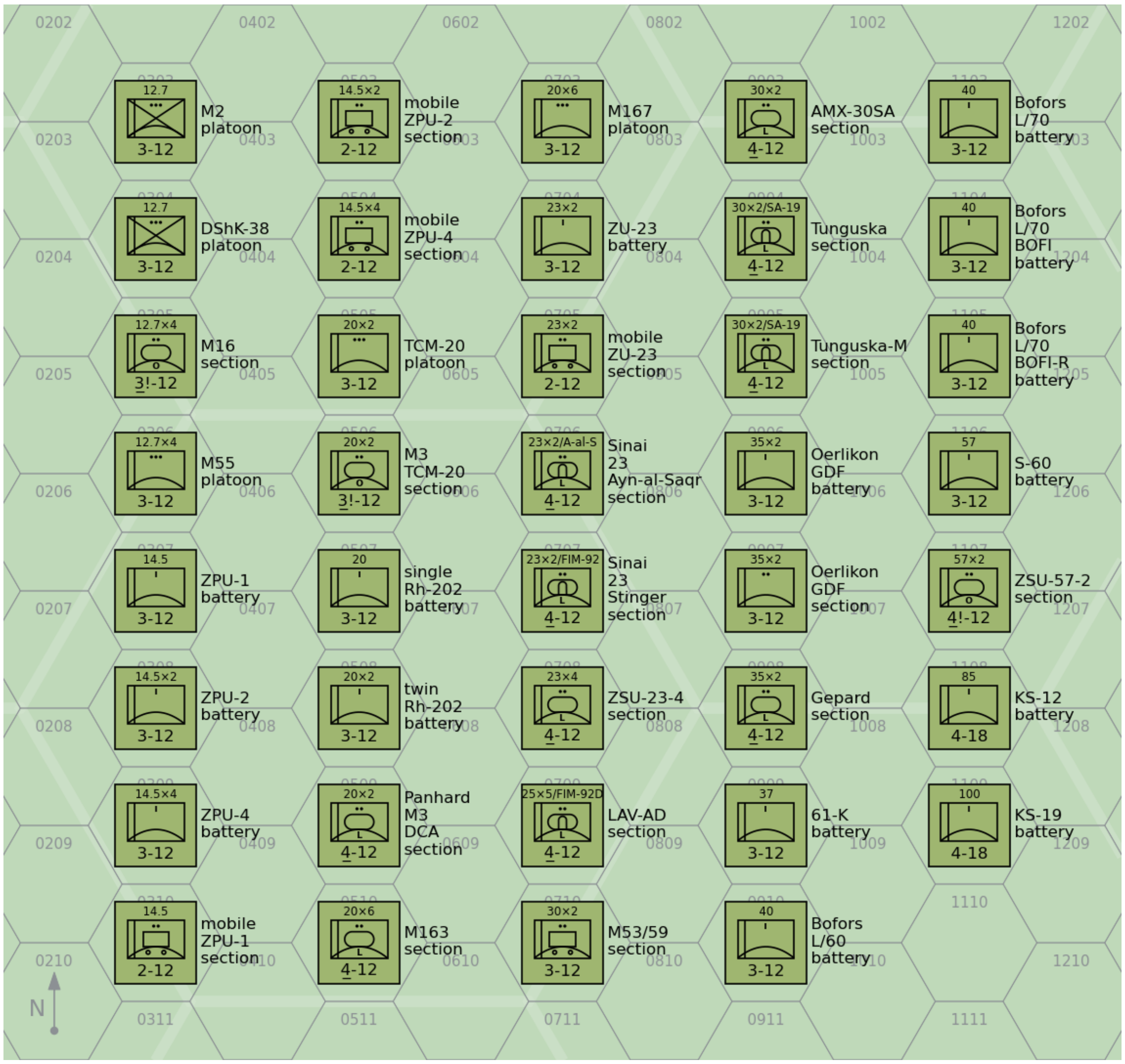


Figure 2: Air-Defense Units: AAA Units

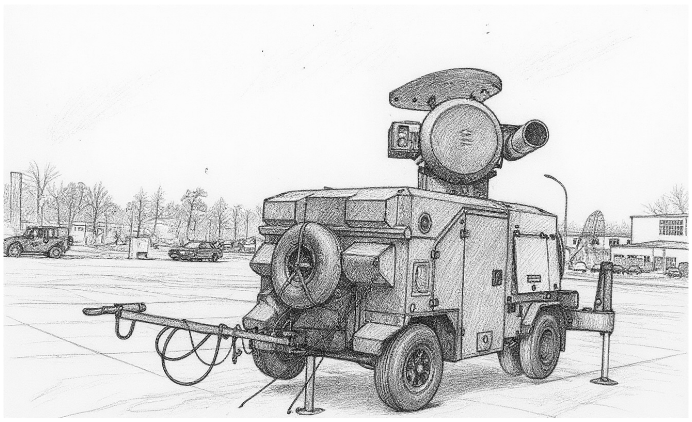
Table 3: Air-Defense Units: AAA Units

Type	Size[a]	Year	Defense Strength	Sighting Range	Mobility	VPs	Gun	AAA Class	Altitude	Range			Hit			Damage Rating	FCR	Night IR Sights	SAM
						3D/2D/D				S	M	L	S	M	L				
M2	Platoon	1933	3	12	U	3/2/1	12.7 mm	L	5	2	3	4	3	2	1	1	—	—	—
DShK-38	Platoon	1938	3	12	U	3/2/1	12.7 mm	L	5	2	3	4	3	2	1	1	—	—	—
M16	Section	1944	<u>3</u>!	12	U	6/3	12.7 mm × 4	L	5	2	3	4	4	3	2	2	—	—	—
M55	Platoon	1945	3	12	T	4/3/1	12.7 mm × 4	L	5	2	3	4	5	4	3	2	—	—	—
ZPU-1	Battery	1949	3	12	T	3/2/1	14.5 mm	L	6	2	3	5	3	2	1	1	—	—	—
ZPU-2	Battery	1949	3	12	T	4/3/1	14.5 mm × 2	L	6	2	3	5	4	3	2	1	—	—	—
ZPU-4	Battery	1949	3	12	T	5/3/2	14.5 mm × 4	L	6	2	3	5	5	4	3	2	—	—	—
Mobile ZPU-1	Section	1949	2	12	G/P	2/1	14.5 mm	L	6	2	3	5	2	1	0	1	—	—	—
Mobile ZPU-2	Section	1949	2	12	G/P	3/2	14.5 mm × 2	L	6	2	3	5	3	2	1	1	—	—	—
Mobile ZPU-4	Section	1949	2	12	G/P	4/2	14.5 mm × 4	L	6	2	3	5	4	3	2	2	—	—	—
M167	Platoon	1967	3	12	T	7/5/2	20 mm × 6	L	8	2	4	6	6	5	4	3	R/HF	—	—
M163	Section	1968	<u>4</u>	12	U	8/4	20 mm × 6	L	8	2	4	6	5	4	3	3	R/HF	—	—
TCM-20	Platoon	1970	3	12	T	5/3/2	20 mm × 2	L	8	2	4	6	4	4	3	2	—	—	—
M3 TCM-20	Section	1970	<u>3</u>!	12	U	6/3	20 mm × 2	L	8	2	4	6	3	3	2	2	—	—	—
Single Rh-202	Battery	1972	3	12	T	4/3/1	20 mm	L	8	2	4	6	4	3	2	2	—	—	—
Twin Rh-202	Battery	1972	3	12	T	5/3/2	20 mm × 2	L	8	2	4	6	5	4	3	3	—	—	—
Panhard M3 DCA	Section	1975	<u>4</u>	12	U	7/4	20 mm × 2	L	8	2	4	6	4	4	3	2	R/UF	—	—
ZU-23	Battery	1960	3	12	T	6/4/2	23 mm × 2	L	9	2	5	8	5	4	3	3	—	—	—
Mobile ZU-23	Section	1960	2	12	G/P	5/3	23 mm × 2	L	9	2	5	8	4	3	2	3	—	—	—
Sinai 23 Ayn-al-Saqr	Section	1992	<u>4</u>	12	U	10/5	23 mm × 2	L	9	2	5	8	4	3	2	3	—	Y	Ayn-al-Saqr
Sinai 23 Stinger	Section	1992	<u>4</u>	12	U	12/6	23 mm × 2	L	9	2	5	8	4	3	2	3	—	Y	FIM-92
ZSU-23-4	Section	1965	<u>4</u>	12	U	9/5	23 mm × 4	L	9	2	5	8	5	4	3	3	W/VF	—	—
LAV-AD	Section	1997	<u>4</u>	12	U	14/7	25 mm × 5	L	10	3	6	9	4	3	2	4	—	Y	FIM-92D
M53/59	Section	1959	3	12	G/P	7/4	30 mm × 2	L	11	4	7	9	3	2	1	3	—	—	—
AMX-30SA	Section	1979	<u>4</u>	12	U	10/5	30 mm × 2	L	11	4	7	9	5	4	3	3	W/VF	—	—
Tunguska	Section	1984	<u>4</u>	12	U	20/10	30 mm × 2	L	11	4	7	9	5	4	3	4	W/VF	—	SA-19
Tunguska-M	Section	1990	<u>4</u>	12	U	22/11	30 mm × 2	L	11	4	7	9	5	4	3	4	W/VF	—	SA-19
Oerlikon GDF	Battery	1963	3	12	T	7/5/2	35 mm × 2	M	12	4	8	10	4	3	2	4	—	—	—
Oerlikon GDF	Section	1963	3	12	T	5/2	35 mm × 2	M	12	4	8	10	3	2	1	4	—	—	—
Gepard	Section	1976	<u>4</u>	12	U	11/6	35 mm × 2	M	12	4	8	10	5	4	3	4	W/VF	—	—
61-K	Battery	1939	3	12	T	5/3/2	37 mm	M	12	4	8	11	3	2	1	4	—	—	—
Bofors L/60	Battery	1935	3	12	T	5/3/2	40 mm	M	15	4	8	12	2	2	1	4	—	—	—
Bofors L/70	Battery	1952	3	12	T	6/4/2	40 mm	M	15	4	8	12	3	3	2	4	—	—	—
Bofors L/70 BOFI	Battery		3	12	T	8/5/3	40 mm	M	15	4	8	12	5	5	4	4	—	—	—
Bofors L/70 BOFI-R	Battery		3	12	T	9/6/3	40 mm	M	15	4	8	12	6	6	5	4	W/VF	—	—
S-60	Battery	1950	3	12	T	8/5/3	57 mm	M	18	5	10	15	2	2	1	5	—	—	—
ZSU-57-2	Section	1955	<u>4</u>!	12	U	6/3	57 mm × 2	M	18	5	10	15	2	1	1	5	—	—	—
KS-12	Battery	1939	4	18	T	9/6/3	85 mm	H	27	6	12	18	2	1	0	6	—	—	—
KS-19	Battery	1948	4	18	T	10/7/3	100 mm	H	39	7	14	21	1	1	0	6	—	—	—

[a] Platoons and batteries are available as sections with corresponding modifications to their damage resiliences and hit rolls.

FCR UNITS

Most medium and heavy AAA batteries can be associated with an add-on fire-control radar (FCR), which improves their accuracy and allows them to engage unsighted targets at night and in adverse weather.



Properties: Table 4 summarizes the properties of FCR units, and Figure 3 shows their representation. In addition to the properties common to other ground units, Table 4 gives:

– the FCR class;
– the frequency band; and
– its modifier to the hit roll.

Specific FCRs:

SON-9: The Soviet SON-9 (Fire Can) FCR-A was derived from the earlier SON-4, which in turn was derived from the US-supplied SCR-584. It is commonly used with 57 mm S-60, 85 mm KS-12, 100 mm KS-19, and 130 mm KS-30 batteries.[68]

The SON-9 entered service in 1955 with Soviet forces and was widely exported to Soviet allies. It has seen combat, including with North Vietnam in the Vietnam War, Egypt and Syria in the 1967 War, the War of Attrition, and the 1973 War.

Super Fledermaus: The Contraves Super Fledermaus FCR-C can be used to control a section of two Oerlikon GDF 35 mm guns; it cannot control a whole battery.[69]

It entered service in 1965 with the Swiss Air Force and also was used by the Argentine Air Force, Brazil, Iran, Japan, and South Africa. It saw combat in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport).

Flycatcher: The Signaal Flycatcher FCR-C can control three to six Bofors L/60 or L/70 guns or SAM launchers.

Flycatcher entered service with the Royal Netherlands Air Force in 1979. It subsequently served with the Indian Armed Forces from 1985, the Irish Defense Forces from 2003, the Royal Netherlands Army from 1987,the Thai Armed Forces from 1981, Turkey, and Venezuela.

Skyguard: The Contraves Skyguard FCR-D can control two Oerlikon GDF 35 mm guns and additionally a SAM launcher with AIM-9 Sparrow, RIM-9 Sea Sparrow, or Aspide missiles.

Skyguard entered service in 1977 with the Swiss Air Force. It was subsequently very widely exported and used by the Argentine Army, Austria, Bahrain, Bangladesh, Brazil, Canada, Chile, China, Cyprus, Egypt, Greece, Indonesia, Iran, Italy, South Korea, Kuwait, Malaysia, Oman, Pakistan, Saudi Arabia, Spain, Taiwan, and the UK. It saw combat with the Argentine Army in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).

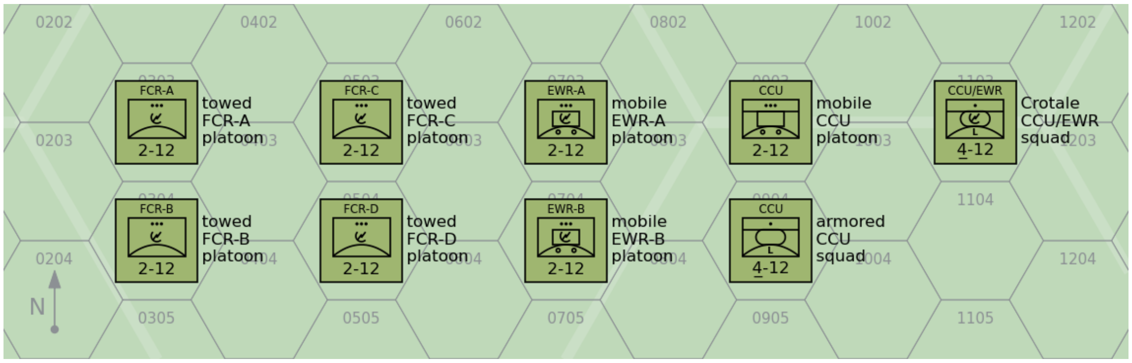


Figure 3: Air-Defense Units: FCR, EWR, and CCU Units

Table 4: Air-Defense Units: FCR Units

Type	Size	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	FCR Class	FCR Frequency	FCR Modifier
Towed FCR-A	Platoon	2	12	T	6/4/2	A	LF	−1
Towed FCR-B	Platoon	2	12	T	6/4/2	B	MF	−2
Towed FCR-C	Platoon	2	12	T	7/5/2	C	VF	−2
Towed FCR-D	Platoon	2	12	T	8/5/3	D	MW	−3

SAM UNITS

SAM launcher units are specialized in the destruction of attacking aircraft with surface-to-air missiles.

Properties: Table 5 summarizes the properties of infantry SAM units, and Figure 5 shows their representation as counters. In addition to the properties shared by all ground units, Table 5 gives:

– the number of ready, volley, and reload missiles;
– the multi-target capability;
– whether the unit has quick-reaction capability;
– whether the unit has IFF capability;
– whether the unit has night IR sights; and
– the optical lock-on number and range.

Missile Properties: Table 7 summarizes the properties of the actual missiles:

– guidance mode;
– launch roll;
– turn rate;
– flight time;
– visibility;
– ECCM, chaff, and flare ratings;
– whether the missile is active homing (AH) or has a home-on-jam (HOJ) capability;
– boost phase in FPs;
– base speed and sustainer duration in game turns;
– minimum altitude (and modifier for targets at T level);
– hit rolls for direct and proximity hits; and
– damage ratings for direct and proximity hits.

Infantry SAM Units: Infantry SAM units are squads of infantry specialized in the destruction of attacking aircraft with portable SAMs. Each squad has one launcher and, unless otherwise specified, one ready missile and two reload missiles.

Infantry SAM squads may be attached to infantry, infantry weapons, or infantry HQ platoons.

Most infantry SAMs do not have true proximity fuzes. Thus, proximity hits in the game correspond to glancing direct hits in reality.

SA-7A/B Squad: The 9K32 Strela-2 (SA-7A Grail) was the first Soviet portable SAM. It has a rear-aspect, uncooled IR seeker.

The 9K32M Strela-2M (SA-7B Grail) is similar, but uses a rear-aspect, electrically-cooled IR seeker. In Warsaw Pact armies, the Strela-2M was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions.[70][71]

The Strela-2 and -2M entered service with Soviet forces in 1968 and 1970. They have seen combat in almost every conflict since 1970 in which the Soviet Union or its successor states have been participants or supporters. In particular, the Strela-2M was used heavily by North Vietnamese forces in the 1973 Easter Offensive, by Arab forces in the 1973 Yom Kippur War, by Angolan forces in the South African Border War, by the Afghan Mujahedeen in the Soviet-Afghan War (having been supplied by the CIA), by Iran in the Iran-Iraq War in the 1980s, and by Serbian and Serbian-supported forces in the Yugoslav Wars in the 1990s.

SA-14 Squad: The 9K34 Strela-3 (SA-14 Gremlin) is a development of the earlier Strela-2M. It has a rear-aspect, cryogen-cooled IR seeker. In Warsaw Pact armies, the Strela-3 was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions.[72]

The Strela-3 entered service with the Soviet Army in 1974. It has seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. In particular, it was used by Iraqi forces in the 1991 Gulf War.

SA-16/18 Squad: 9K310 Igla-1 (SA-16 Gimlet) and 9K38 Igla (SA-18 Grouse) use very similar missiles and launchers, but the Igla-1 has a single-channel IR seeker similar to the SA-14, and the Igla uses a dual-channel IR seeker to reduce its susceptibility to flares. The warhead was similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). Both incorporate IFF interrogators, but these were only used in Warsaw Pact armies.[73]

The Igla-1 entered service with the Soviet Army in 1981 and the Igla in 1983. Both have been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. The Igla was used by Iraqi forces in the 1991 Gulf War, by the Peruvian Army in the 1995 Cenepa War, and by Serbian forces in the 1999 Kosovo War.

FIM-43C Squad: The FIM-43C Redeye Block III was the first US infantry SAM to reach production and enter service. It has a rear-aspect, cryogen-cooled IR seeker.[74]

The FIM-43C entered service with the US Army in 1968 and subsequently was used by Australia, Chad, Denmark, West Germany, Greece, Israel, Jordan, Saudi Arabia, Somalia, Sudan, Sweden, and Turkey. It was also supplied by the CIA to the Afghan Mujahedeen and the Nicaraguan Contras.

FIM-92A/B/C/D Squad: The FIM-92A Stinger has an all-angle, second-generation, cryogen-cooled IR seeker. Its warhead weight 3 kg and was significantly more lethal than in earlier missiles. The FIM-92B Stinger POST adds a UV seeker to distinguish flares from aircraft. The FIM-92C/D Stinger RMP have software updates to further improve the performance against flares. All have an IFF interrogator to help avoid fratricide.[75]

The FIM-92A entered service with the US Army in 1981, the

FIM-92B in 1986, the FIM-92C in 1989, and the FIM-92D in 1992. They were exported to dozens of US allies including the Afghan Mujahedeen and the Angolan UNITA. They saw service in the 1982 South Atlantic War, the Soviet-Afghan War, the Angolan Civil War, the Libyan Invasion of Chad, the Tajik Civil War, the Chechen War, the Sri Lankan Civil War, the Syrian Civil War, and the Russian Invasion of Ukraine.

Blowpipe Squad: Blowpipe is an MCLOS optically-guided missile. While in theory this gave it an all-aspect capability, in practice guiding the missile to the target in combat proved to be very difficult.[76]

Blowpipe entered service with the British Army and Royal Marines in 1975. It was also used by the Argentine Army, Canadian Army, Chilean Army and Navy, Ecuadorean Army, Guatemalan Army, Portuguese Army, Malawian Army, Malaysian Army, Nigerian Army, Omani Army, Royal Thai Air Force and Army, and the UAE Army. It saw combat in the 1982 South Atlantic War with both sides and had a low success rate. In 1986, it was trialed by the Afghan Mujahedeen, but again was not considered to be effective. Finally, it was also used in the 1995 Cenepa War by Ecuador.

Javelin/Starburst Squad: Given the poor performance of Blowpipe in the 1982 South Atlantic War, Javelin was quickly developed as an upgraded version of Blowpipe with SACLOS guidance in place of MCLOS guidance. Starburst then replaced the radio-link with laser guidance. Javelin is also called "Javelin GL" and Starburst was originally called "Javelin S15".[77]

Javelin replaced Blowpipe in the British Army and Royal Marines in 1984. It was also used by the Botswana Army, the Canadian Army, the Malaysian Army, the Peruvian Army, and the South Korean Army.

Starburst in turn replaced Javelin in the British Army and Royal Marines in 1989. It was also used by the Canadian Army, the Malaysian Army, the Qatari Army, and the Royal Thai Army. Starburst served with the British Army in the 1991 Gulf War, but was not used in combat.

Starstreak Squad: Starstreak is a high-speed laser-guided missile. As the missile approaches the target, it launches three independent explosive darts. The darts have contact fuzes, but not proximity fuzes.[78]

Starstreak replaced Javelin in the British Army and Royal Marines in 2000 and is also used by the South African Army and the Armed Forces of Ukraine. Starstreak served with the British Army in the 2003 Invasion of Iraq, but was not used in combat. It has seen combat in the Russian Invasion of Ukraine.

Starstreak LML Squad: The Starstreak Light Multiple Launcher (LML) has three ready-to-fire Starstreak missiles on a pedestal mount with a single sighting unit. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit.[79]

Starstreak LML entered service with the British Army and Royal Marines in 2000. It is also used by Indonesia, Malaysia, and South Africa.

Mistral Squad: The Mistral is a French IR-homing missile on a pedestal mount. The missile has a considerably larger warhead than either the Stinger or Igla missiles, but this comes at a cost in portability. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit.[80]

The Mistral entered service with the French Army in 1990. It was subsequently exported to many other countries.

RBS 70 Squad: The RBS 70 is a Swedish laser-guided missile on a pedestal mount. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit.[81]

The RBS 70 entered service with the Swedish Army in 1977. It was subsequently used by Argentina, Australia, Bahrain, Brazil, the Czech Republic, Iran, Ireland, Lithuania, Norway, Pakistan, Singapore, Thailand, Tunisia, the UAE, Ukraine, and Venezuela.

Ayn-al-Saqr Squad: The Ayn-al-Saqr ("Eye of the Hawk") missile is a licensed version of the 9M32M Strela-2m (SA-7B Grail) manufactured in Egypt. It equips one version of the Sinai 23 air-defense vehicle.[82]

It entered service with the Egyptian Army in 1984.

Vehicular and Towed SAM Units:

Echelon: The size of a SAM launcher unit is squad, section, or platoon according to its damage resilience (1D, 2D, or 3D, respectively). This does not always coincide with the actual designations used historically. For example, the SA-2 battery in the game corresponds to a battalion in reality.

Nuclear Warheads: The SA-2E has the option of a nuclear warhead.

Soviet and Russian SAM Launcher Units:

SA-2B/C/E/F Battery: This is an S-75 (SA-2 Guideline) battery of six single launchers for the long-range V-750 missile and their associated P-12 (Spoon Rest) EWR and SNR-75 (Fan Song) TTR. The variants are the S-75 Desna (SA-2B), S-75M Volkhov (SA-2C), S-75AK (SA-2E), and S-75SM (SA-2F). All are radar-guided, but the SA-2F also has an optical backup.

The S-75 Dvina (SA-2A) system was deployed by the Soviet PVO in 1957. In addition to serving as a key part of the air-defense system of the Soviet Union (and its notable role in ending reconnaissance overflights by shooting down the U-2 of Gary Powers) and its allies, it saw combat in Cuba in 1962, with North Vietnamese forces from 1965, with Indian forces in the 1965 war with Pakistan, and with Egyptian and Syrian forces from 1967.

SA-3B Battery: This is an S-125 Neva (SA-3B Goa) battery with four quad launchers for the 5V24 missile and their associated P-15 (Flat Face) EWR and SNR-125 (Low Blow) TTR.

It saw combat with Egyptian and Syrian forces from 1970.

Mobile SA-3B Battery: This is a mobile S-125 Neva (SA-3B Goa) battery with four twin launchers for the 5V24 missile on trucks and their associated P-15 (Flat Face) EWR and SNR-125 (Low Blow) TTR.

It saw combat with Egyptian and Syrian forces from 1970 and with Iraqi forces in the Iran-Iraq War, the Gulf War, and the Invasion of Iraq.

SA-4A/B Battery: This is an armored 2K11 Krug (SA-4A/B Ganef) battery with three launcher vehicles each with two long-range 9M8M1/9M8M2 missiles and their associated P-40 (Long Track) EWR and 1S32 (Pat Hand) TTR.

It saw combat with Egyptian and Syrian forces from 1970 and with Iraqi forces in the Iran-Iraq War, the Gulf War, and the Invasion of Iraq.

SA-6A/B Battery: This is an armored 2K12 Kub (SA-6A/B Gainful) battery with four launcher vehicles, each with three medium-range 2K12 Kub (SA-6A) or 3M9M3 (SA-6B) missiles and their associated P-40 (Long Track) EWR and 1S32 (Pat Hand) TTR.

It saw combat with Egyptian and Syrian forces from 1970 and with Iraqi forces in the Iran-Iraq War, the Gulf War, and the Inva-

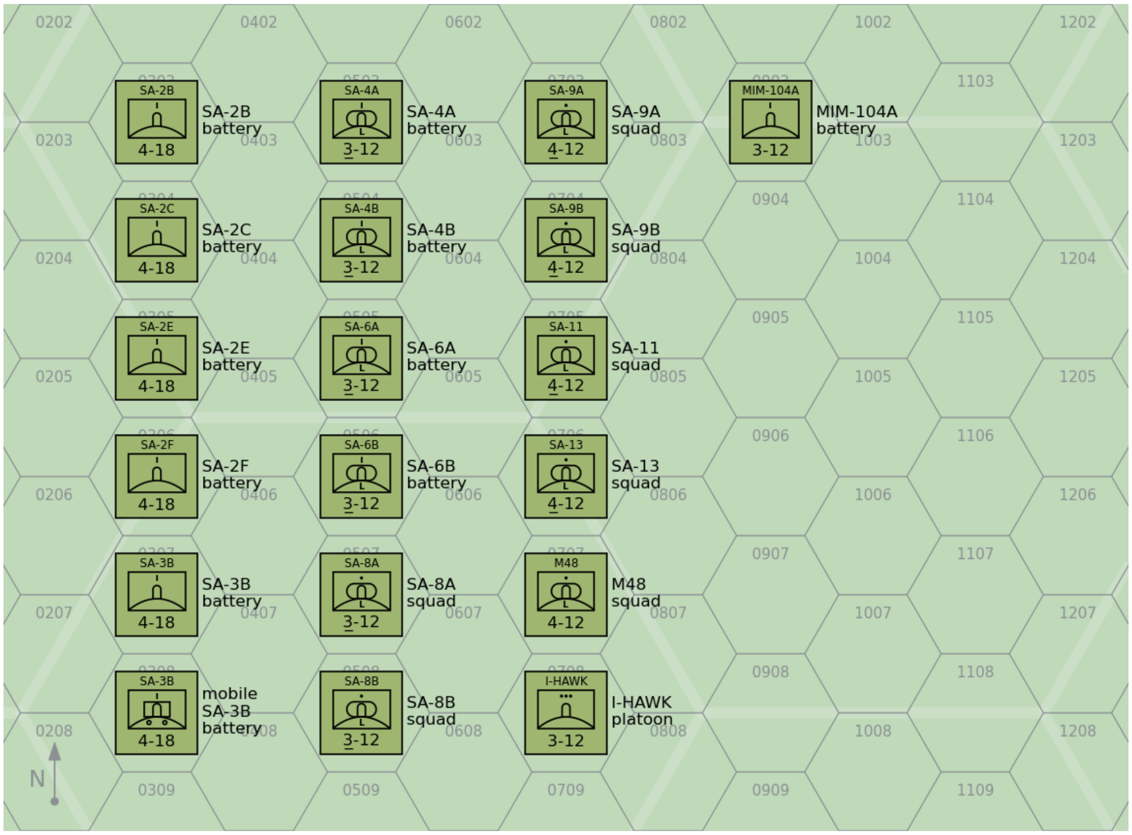


Figure 4: SAM Launcher Units

sion of Iraq.

SA-8A/B Squad: This is a squad with a single 9K33 Osa or 9K33M Osa-AK (SA-8A/B Gecko) vehicle with integrated 1S51M3 (Land Roll) TTR. The SA-8A carries four 9M33 missiles, and the SA-8B carries six 9M33M missiles.

The Osa has been widely exported and seen combat with Syrian forces in the 1982 Lebanon War, with Cuban forces in the South African Border War, with Iraqi forces in the 1990 Gulf War, and in more recent conflicts.

SA-9A/B Squad: This is a squad with a single 9K21 Strela-1 or 9K21M Strela-1M (SA-9A/B Gaskin) vehicle with four 9M21 or 9M21M missiles. In the Soviet army, a platoon of four was assigned to the air-defense battery of tank and motor-rifle regiments, complementing the ZSU-23-4.

The Strela-1 has been widely exported and saw combat in the South African Border War, the Iran-Iraq War, the Lebanon War, the Gulf War, the Kosovo War, and the Invasion of Iraq.

SA-11 Squad: This is a squad with a single 9K37 Buk (SA-11 Gadfly) vehicle with four 9M37 missiles and an integrated 9S35 (Fire Dome) TTR. Three squads make up a battery, and three batteries make up a battalion. Normally, one armored 9S18 (Tube Arm) or 9S18M1 (Snow Drift) EWR is allocated to each battalion.

The SA-11 has seen combat in the War in Abkhazia, the Russo-Ukrainian War, and during the Syrian Civil War.

SA-13 Squad: This is a single 9K35 Strela-10 (SA-13 Gopher) vehicle with four infra-red homing 9M37 missiles. Four squads form a platoon. It began to replace the SA-9B in the air-defense platoons of tank and motor-rifle regiments in the Soviet Army starting in 1976 and was subsequently widely exported to Soviet allies. Compared to the earlier system, the missile offers greater lethality, and the vehicle greater protection and mobility (although it is not amphibious). Some are equipped with 9S16 (Flat Box-B) RWR.

The SA-13 saw combat in the South African Border War with Angolan forces, the Gulf War with Iraqi forces, and in the Kosovo War with Serbian forces.

US SAM Launcher Units:

M48 Squad: This is a single M48 Chaparral vehicle with four adapted Sidewinder infrared-homing missiles. It was developed after the ambitious Mauler SAM program was abandoned and introduced in 1969. Night IR sights were added in 1984 to give it a limited night capability. The MIM-72A was the original missile based on the AIM-9D. The MIM-72C added an improved seeker and warhead, and the MIM-72E/F added a smokeless motor. The final MIM-72G added a seeker based on that of the FIM-92B Stinger, with much improved rejection of infrared countermeasures.

I-HAWK Platoon: This consists of three triple launchers for the MIM-23A/C/D missile, plus supporting radars and command elements. The MIM-23B Improved HAWK and its associated improved radars were developed from the original HAWK MIM-23A to give much improved performance at low levels. The MIM-23C has improved ECCM, and the MIM-23D adds a home-on-jam capability. Two platoons with an EWR make up a battery.

The I-HAWK was deployed by the US Army, US Marines, and many US allies.

MIM-104 Battery: An MIM-104 Patriot battery consists of six launchers with associated radar and command elements. The MIM-104A uses the “Standard” missile and was capable only of defense against aircraft. Patriot is used by the US Army and numerous US allies.

Figure 5: Air-Defense Units: Infantry SAM Units

Table 5: Air-Defense Units: Infantry SAM Units

Name	Size	Year	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	SAM	Missiles: Ready	Missiles: Volley	Missiles: Reload	Multi-Target	Quick Reaction	IFF	Night IR Sights	Lock-On: Optical	Lock-On: Range	Other Names
SA-7A	Squad	1968	5	6	U	–/–/3	SA-7A	1	1	2	—	Y	—	—	7	9	9K32 Strela-2 (Grail)
SA-7B	Squad	1972	5	6	U	–/–/4	SA-7B	1	1	2	—	Y	—	—	7	9	9K32M Strela-2M (Grail)
SA-14	Squad	1974	5	6	U	–/–/6	SA-14	1	1	2	—	Y	—	—	7	12	9K34 Strela-3 (Gremlin)
SA-16	Squad	1981	5	6	U	–/–/8	SA-16	1	1	2	—	Y	Y	—	7	12	9K310 Igla-1 (Gimlet)
SA-18	Squad	1983	5	6	U	–/–/10	SA-18	1	1	2	—	Y	Y	—	8	16	9K38 Igla (Grouse)
FIM-43C	Squad	1968	5	6	U	–/–/4	FIM-43C	1	1	2	—	Y	—	—	7	9	Redeye Block III
FIM-92A	Squad	1981	5	6	U	–/–/9	FIM-92A	1	1	2	—	Y	Y	—	8	12	Stinger
FIM-92B	Squad	1986	5	6	U	–/–/10	FIM-92B	1	1	2	—	Y	Y	—	8	12	Stinger POST
FIM-92C	Squad	1989	5	6	U	–/–/10	FIM-92C	1	1	2	—	Y	Y	—	8	12	Stinger RMP
FIM-92D	Squad	1992	5	6	U	–/–/10	FIM-92D	1	1	2	—	Y	Y	—	8	12	Stinger RMP
Blowpipe	Squad	1975	5	6	U	–/–/6	Blowpipe	1	1	2	—	Y	Y	—	7	9	
Javelin	Squad	1984	5	6	U	–/–/8	Javelin	1	1	2	—	Y	Y	—	7	9	Javelin GL
Starburst	Squad	1989	5	6	U	–/–/8	Starburst	1	1	2	—	Y	Y	—	7	9	Javelin S15
Starstreak	Squad	2000	5	6	U	–/–/12	Starstreak	1	1	2	—	Y	Y	—	7	12	
Starstreak LML	Squad	2000	4	6	U	–/–/14	Starstreak	3	1	0	—	Y	Y	—	7	12	
Mistral	Squad	1990	4	6	U	–/–/11	Mistral	1	1	2	—	Y	—	—	7	12	
RBS 70	Squad	1977	4	6	U	–/–/8	RBS 70	1	1	2	—	Y	Y	—	7	10	
Ayn-al-Saqr	Squad	1984	5	6	U	–/–/4	Ayn-al-Saqr	1	1	2	—	Y	—	—	7	12	

Table 6: Air-Defense Units: SAM Launcher Units

						VPs		EWR			TTR		Missiles						Lock-On			
Name	Size	Year	Defense Strength	Sighting Range	Mobility	3D/2D/D	SAM	Frequency	Range	MTI	Frequency	Range	Ready	Volley	Reload	Multi-Target	Quick Reaction	Night IR Sights	Radar	Optical	Range	Other Names
SA-2B	Battery	1957	4	18	G/P	8/5/3	SA-2B	LF	30	—	LF	20	6	2	0	—	—	—	6	—	—	S-75 Desna (Guideline)
SA-2C	Battery	1960	4	18	G/P	8/5/3	SA-2C	LF	30	—	MF	20	6	2	0	—	—	—	6	—	—	S-75M Volkhov (Guideline)
SA-2E	Battery	1962	4	18	G/P	8/5/3	SA-2E	LF	30	—	MF	20	6	2	0	—	—	—	6	—	—	S-75AK (Guideline)
SA-2F	Battery	1967	4	18	G/P	8/5/3	SA-2F	LF	30	—	LF	20	6	2	0	—	—	—	6	6	—	S-75SM (Guideline)
SA-3B	Battery	1964	4	18	G/P	10/7/3	SA-3B	MF	35	—	VF	10	16	2	0	—	—	—	7	6	—	S-125M Neva-M (Goa)
Mobile SA-3B	Battery	1964	4	18	G/P	12/8/4	SA-3B	MF	35	—	VF	10	16	2	0	—	—	—	7	6	—	S-125M Neva-M (Goa)
SA-4A	Battery	1963	3	12	U	10/7/3	SA-4A	VF	50	—	HF	30	6	2	0	—	—	—	7	—	—	2K11 Krug (Ganef)
SA-4B	Battery	1973	3	12	U	10/7/3	SA-4B	VF	50	—	HF	30	6	2	0	—	—	—	7	—	—	2K11M Krug-M (Ganef)
SA-6A	Battery	1966	3	12	U	11/7/4	SA-6A	MF	24	—	HF	18	12	2	0	—	—	—	7	6	—	2K12 Kub (Gainful)
SA-6B	Battery	1977	3	12	U	12/8/4	SA-6B	MF	24	—	HF	18	12	2	0	—	—	—	7	6	—	2K12M3 Kub-M3 (Gainful)
SA-8A	Squad	1967	3	12	U	–/–/10	SA-8A	HF	10	—	VF	6	4	2	0	2	Y	—	8	—	—	9K33 Osa (Gecko)
SA-8B	Squad	1980	3	12	U	–/–/10	SA-8B	HF	10	—	VF	6	6	2	0	2	Y	—	8	7	—	9K33M2 Osa-AK (Gecko)
SA-9A	Squad	1968	4	12	U	–/–/7	SA-9A	—	—	—	—	—	4	2	0	2	Y	—	—	7	12	9K31 Strela-1 (Gaskin)
SA-9B	Squad	1970	4	12	U	–/–/7	SA-9B	—	—	—	—	—	4	2	0	2	Y	—	—	7	12	9K31M Strela-1M (Gaskin)
SA-11	Squad	1980	4	12	U	–/–/12	SA-11	—	—	—	HF	18	4	2	0	2	—	—	8	7	—	9K37 Buk (Gadfly)
SA-13	Squad	1976	4	12	U	–/–/10	SA-13	—	—	—	—	—	4	2	0	2	Y	—	—	8	12	9K35 Strela-10 (Gopher)
M48	Squad	1969	4	12	U	–/–/9	MIM-72	—	—	—	—	—	4	2	0	2	Y	Y	—	7	12	Chaparral
I-Hawk	Platoon	1971	3	12	G/P	12/8/4	MIM-23B	—	—	—	VF	20	9	2	0	—	Y	—	7	7	—	
MIM-104A	Battery	1984	3	12	G/P	45/30/15	MIM-104A	LF	60	Y	LF	40	24	8	0	8	Y	—	9	—	—	Patriot (Standard)

Table 7: SAMs

																	Hit		Damage		
Name	Year	Guidance Mode	Launch Roll	Turn Rate	Flight Time	Visibility	ECCM	Chaff	Flare	AH	HOJ	Instant Arming	Boost Phase	Base Speed	Sustainer	Minimum Altitude	Direct	Proximity	Direct	Proximity	Other Names
SA-7A	1968	I	6	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	6	4	2	9M32 Strela-2 (Grail)
SA-7B	1972	I	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	6	4	2	9M32M Strela-2M (Grail)
SA-14	1974	M	7	BT	1	6	—	—	4	—	—	Y	—	9	0	T	6	7	4	2	9M34 Strela-3 (Gremlin)
SA-16	1981	M	8	BT/2	1	6	—	—	4	—	—	Y	—	12	0	T	6	7	4*	2*	9K310 Igla-1 (Gimlet)
SA-18	1983	A	8	BT/2	1	6	—	—	2	—	—	Y	—	12	0	T	7	8	4*	2*	9K38 Igla (Grouse)
FIM-43C	1968	M	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	6	7	4	2	Redeye Block III
FIM-92A	1981	A	8	BT/2	1	2	—	—	3	—	—	Y	—	14	0	T	7	8	5	3	Stinger
FIM-92B	1986	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger POST
FIM-92C	1989	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger RMP
FIM-92D	1992	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger RMP
Blowpipe	1975	OG	8	HT	1	6	—	3	0	—	—	—	—	8	0	T	3	7	6	5	
Javelin	1984	OG	8	ET	1	5	—	2	0	—	—	Y	—	10	0	T	5	8	6	5	Javelin GL
Starburst	1989	LG	8	ET	1	5	—	1	0	—	—	Y	—	10	0	T	5	8	6	5	Javelin S15
Starstreak	2000	LG	8	ET	1	5	—	1	0	—	—	Y	—	24	0	T	7	9	6	4	
Mistral	1990	A	8	ET	1	5	—	—	2	—	—	Y	—	18	0	T	7	9	6	3	
RBS 70	1977	LG	8	ET	1	6	—	2	0	—	—	—	—	12	0	T	5	8	6	3	
Ayn-al-Saqr	1984	I	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	7	4	3	

EWR AND CCU UNITS

An early-warning radar (EWR) can detect aircraft at longer ranges than the target acquisition radars typically associated with SAM units. It can then pass this information down to SAMs in their integrated air-defense system, sometimes with the intermediation of a CCU, to improve the chance of a target acquisition and lock-on.

A command-and-control unit (CCU) acts to coordinate multiple independent SAM firing units in an integrated air-defense system. For early SAMs, where each battery can engage one target, CCUs are typically at the regimental level (i.e., the formation above the battery). For later SAMs, where each launcher can engage one or more targets, they are typically at the battery level.

Figure 3 shows the representation of EWR and CCU units as counters, and Table 8 summarizes their properties.

Properties: In addition to the properties common to other ground units, Table 8 gives: the EWR class, its frequency band, and its range in mega-hexes.

EWR-A:

EWR-B:

Mobile CCU:

Armored CCU:

Crotale CCU/EWR: This is the HQ and EWR vehicle of a Crotale SAM battery. It is equipped with a Mirador IV radar.

Table 8: Air-Defense Units: EWR and CCU Units

					VPs	EWR		
Type	Size	Defense Strength	Sighting Range	Mobility	3D/2D/D	Class	Frequency	Range
Mobile EWR-A	Platoon	2	12	G/P	12/8/4	A	LF	—
Mobile EWR-B	Platoon	2	12	G/P	12/8/4	B	LF	—
Mobile CCU	Platoon	2	12	G/P	8/5/3	—	—	—
Armored CCU	Squad	$\underline{4}$	12	U	–/–/8	—	—	—
Crotale CCU/EWR	Squad	$\underline{4}$	12	U	–/–/10	B	LF	8

GROUND TARGETS

A ground *target* is a unit of infrastructure such as a bridge and a building.

Properties and Representation: Ground targets are listed with their properties in Table 9.

All ground targets have these properties:

– defense strength;

– defense strength and target class;

– sighting range in hexes; and

– the VPs awarded for 3D/2D/D damage.

All ground targets have a damage resilience of 3D. That is, they are killed when they suffer 3D or more damage.

Some ground targets are represented by counters, and others are represented by features on the maps. Figure 6 shows the graphical representation of the counters, which mainly uses adapted NATO symbols.

Notes on Specific Targets:

Aircraft: No VP values are given for transport or fighter airplanes or helicopters, since these can vary significantly with the capacity of the aircraft. A scenario must specify the VP values. As a guideline, they will often be similar to the VP values of the attacking aircraft.

Bridges: Unless otherwise indicated in a scenario, a major bridge is one that extends over two or more hexes, a minor bridge is one that is confined to a single hex, and a small bridge is where a road crosses a river without a bridge being marked on the map.

Penetrating Bombs: When attacking a bunker entrance, a tunnel entrance hex, or a shelter, the attack strength of a penetrating bomb is doubled.

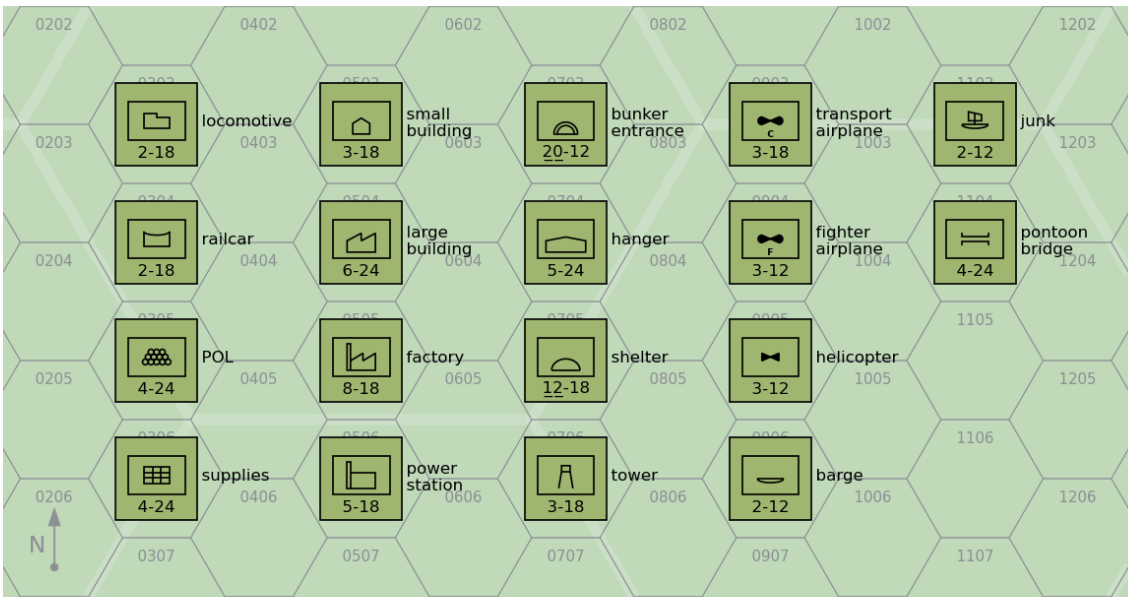


Figure 6: Ground Targets

Table 9: Ground Targets

Type	Counter?	Defense Strength	Sighting Range	VPs 3D/2D/D
Locomotive	Y	2	18	12/8/4
Railcar	Y	2	18	5/3/2
POL	Y	4	24	15/10/5
Supplies	Y	4	24	15/10/5
Small Building	Y	3	18	4/3/1
Large Building	Y	6	24	7/5/2
Factory	Y	8	18	20/13/7
Power Station	Y	5	18	12/8/4
Bunker Entrance	Y	20	12	30/20/10
Barge	Y	2	12	4/3/1
Junk	Y	2	12	4/3/1
Hanger	Y	5	24	8/5/3
Shelter	Y	12	18	12/8/4
Tower	Y	3	18	5/3/2
Transport Airplane	Y	3	18	
Fighter Airplane	Y	3	12	
Helicopter	Y	3	12	
Railroad Hex	N	6	24	5/3/2
Railyard Hex	N	10	24	12/8/4
Docks Hex	N	10	36	8/5/3
Piers Hex	N	10	36	8/5/3
Road Hex	N	8	24	2/1/0
Trail Hex	N	6	12	
Tunnel Entrance Hex	N	12	0	10/7/3
Runway Hex	N	10	36	8/5/3
City Hex	N	5	48	8/5/3
Town Hex	N	3	36	2/1/0
Village Hex	N	3	36	2/1/0
Major Bridge	N	18	48	30/20/10
Minor Bridge	N	12	36	18/12/6
Small Bridge	N	6	24	6/4/2
Pontoon Bridge	Y	4	24	6/4/2
Dam	N	20	24	15/10/5

EXAMPLE ORDERS OF BATTLE

This appendix contains a number of example orders of battle and their correspondence in game units.

– Table 10: 2 Para Battalion at the 1982 Battle of Goose Green. At this time, the battalion had attached half a battery of 105 mm Light Guns from Alma Battery of 29 Commando Regiment, RA, a section of Blowpipes from 43rd Air Defence Battery, RA, and a Recce Troop of 59 Independent Commando Squadron, RE.

– Table 11: 12th Infantry Regiment at the 1982 Battle of Goose Green, with attachments from the 25th Infantry Regiment, 4th Airborne Artillery Regiment, 601st Anti-Aircraft Artillery Group.

– Tables 12, 13, 12, 14, 15, and 16: An armored regiment, APC infantry battalion, infantry battalion, field artillery regiment, and heavy artillery regiment of the British Army from about 1960. In the British Army, regiments are battalion-level units.[83]

– Table 17: An armored regiment of the British Army from about 1985.

– Tables 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, and 31: A Soviet front-line motor-rifle division from the 1980s with its constituent regiments and battalions.[84] In the Soviet Army, regiments are brigade-level units.

Table 10: 2 Para at Goose Green

Element	Game Ground Unit	Notes
Battalion HQ	1 × infantry HQ platoon	
A Company	3 × infantry platoon	
B Company	3 × infantry platoon	
C Company	2 × infantry platoon	Patrol Company
D Company	3 × infantry platoon	
Support Company	1 × infantry mortar platoon	
	1 × infantry weapons platoon	GPMGs
	1 × infantry weapons platoon	MILANs
	1 × infantry platoon	Engineers
Alma Battery	1 × towed artillery section	Three 105-mm Light Guns
Blowpipe Section	2 × infantry SAM squad – Blowpipe	
Engineers Troop	1 × infantry platoon	

Table 11: 12IR at Goose Green

Element	Game Ground Unit	Notes
Regimental HQ	1 × infantry HQ platoon	
SAM Section	2 × infantry SAM squad – SA-7B	
A Company 12IR	5 × infantry platoon	
B Company 12IR	3 × infantry platoon	
C Company 25IR	2 × infantry platoon	
	1 × infantry weapons platoon	
C Company 12IR	2 × infantry platoon	
	1 × infantry weapons platoon	
A Battery, 4AAR	1 × towed artillery section	Three 105-mm Pack Howitzers
B Battery, 601 GADA	1 × Oerlikon 35 mm GDF section	Two guns
	1 × towed FCR-D	Skyguard
9th Engineering Company	1 × infantry platoon	
3 Section B Battery 601AAAG	1 × Rh-202 20 mm battery	Six guns
	1 × EWR	Elta
Security Company	1 × infantry platoon	

Table 12: British Army Armored Regiment (Early 1960s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon	
1 × Reconnaissance Squadron	3 × light reconnaissance/tank platoon	15 × Ferret/Saladin
3 × Tank Squadrons (A/B/C) each:	3 × medium tank platoon	14 × Centurion
	1 × heavy tank platoon	3 × Conqueror

Table 13: British Army APC Infantry Battalion (Early 1960s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
3 × Infantry Companies (A/B/C) each:	3 × infantry platoon	
	1 × infantry weapons platoon	
	4 × light APC platoon	17 × Saracen/FV432

Table 14: British Army Infantry Battalion (Early 1960s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × infantry HQ platoon	
	1 × truck platoon	
4 × Infantry Companies (A/B/C/D) each:	3 × infantry platoon	
	1 × infantry weapons platoon	
	4 × truck platoon	

Table 15: British Army Field Artillery Regiment (Early 1960s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × mobile HQ platoon	
4 × Batteries (A/B/C/D) each:	1 × tracked artillery battery	6 × Cardinal (M44)

Table 16: British Army Heavy Artillery Regiment (Early 1960s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × mobile HQ platoon	
2 × Batteries (A/B) each:	1 × mobile rocket section	2 × Honest John
2 × Batteries (C/D) each:	1 × towed artillery battery	4 × 8-inch Gun M1
	1 × truck platoon	

Table 17: British Army Armored Regiment (Mid 1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon	
1 × Reconnaissance Troop	2 × light tank platoon	8 × Scorpion
1 × ATGW Troop	3 × light anti-tank platoon	9 × FV438
3 × Tank Squadrons (A/B/C) each:	4 × heavy tank platoon	14 × Chieftain

Table 18: Soviet Tank Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
3 × Tank Companies each:	3 × heavy tank platoon	12 × T-64/72/80

Table 19: Soviet BMP Motor Rifle Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
1 × Mortar Battery	1 × towed mortar platoon	M1943 120 mm mortar
	1 × truck platoon	
1 × Grenade Launcher Platoon	1 × light IFV/APC platoon	3 × BMP-1/2
	1 × infantry weapons platoon	6 × AGS-17 grenade launchers
3 × Motor Rifle Companies each:	4 × light IFV platoon	12 × BMP-1/2
	3 × infantry platoon	
	1 × infantry weapons platoon	6 × PKM machine guns
	3 × SA-7/14/16 squad	3 × SA-7/14/16

Table 20: Soviet BTR Motor Rifle Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
1 × Mortar Battery	2 × towed mortar platoon	8 × M1943 mortar
	2 × truck platoon	
1 × Anti-Tank Platoon	1 × light IFV/APC platoon	5 × BTR-60/70
	1 × infantry weapons platoon	2 × recoilless rifle and 4 × ATGM
1 × Grenade Launcher Platoon	1 × light IFV/APC platoon	3 × BTR-60/70
	1 × infantry weapons platoon	6 × AGS-17 grenade launchers
3 × Motor Rifle Companies each:	4 × light APC platoon	12 × BTR-60/70
	3 × infantry platoon	
	1 × infantry weapons platoon	3 × PKM machine gun and 3 × ATGM
	3 × SA-7/14/16 squad	3 × SA-7/14/16

Table 21: Soviet Self-Propelled Artillery Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × mobile HQ platoon	
3 × Howitzer Battery each	1 × towed artillery battery	6 × 122-mm 2S1 Gvozdika

Table 22: Soviet Artillery Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × mobile HQ platoon	
3 × Howitzer Battery each	1 × towed artillery battery	6 × 122-mm D-30
	1 × truck platoon	

Table 23: Soviet Tank Regiment (1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon	
	3 × SA-7/14/16 squad	
	3 × light truck squad	
1 × Reconnaissance Company	1 × light IFV platoon	3 × BMP-1/2
	1 × light reconnaissance platoon	4 × BRDM-2
1 × Air-Defense Battery	2 × SA-9/13 section	4 × SA-9/13
	2 × ZSU-23-4 section	4 × ZSU-23-4
3 × Tank Battalion		
1 × Self-Propelled Artillery Battalion		

Table 24: Soviet BMP Motor-Rifle Regiment (1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon	
	3 × SA-7/14/16 squad	
	3 × light truck squad	
1 × Reconnaissance Company	1 × light IFV platoon	3 × BMP-1/2
	1 × light reconnaissance platoon	4 × BRDM-2
1 × Anti-Tank Battery	3 × light anti-tank platoon	9 × BRDM-2 ATGM
1 × Air-Defense Battery	2 × SA-9/13 section	4 × SA-9/13
	2 × ZSU-23-4 section	4 × ZSU-23-4
3 × BMP Motor-Rifle Battalions		
1 × Tank Battalion		
1 × Self-Propelled Artillery Battalion		

Table 25: Soviet BTR Motor-Rifle Regiment (1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon	
	3 × SA-7/14/16 squad	
	3 × light truck squad	
1 × Reconnaissance Company	1 × light IFV platoon	3 × BMP-1/2
	1 × light reconnaissance platoon	4 × BRDM-2
1 × Anti-Tank Battery	3 × light anti-tank platoon	9 × BRDM-2 ATGM
1 × Air-Defense Battery	2 × SA-9/13 section	4 × SA-9/13
	2 × ZSU-23-4 section	4 × ZSU-23-4
3 × BTR Motor-Rifle Battalions		
1 × Tank Battalion		
1 × Artillery Battalion		

Table 26: Soviet Artillery Regiment (1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon	
3 × Self-Propelled Artillery Battalions each	3 × light armored artillery battery	18 × 152-mm 2S3
1 × Rocket Artillery Battalion	3 × mobile rocket artillery battery	18 × BM-21

Table 27: Soviet SAM Regiment (1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon	
	3 × SA-7/14/16 squad	
	3 × light truck squad	
1 × EWR Battery	2 × EWR-B	2 × Long Track EWR and 1 × Thin Skin HFR
5 × SAM Batteries each	1 × SA-6 battery	4 × SA-6 TELAR and 1 × Straight Flush TTR
	1 × SA-7/14/16 squad	
	1 × light truck squad	

Table 28: Soviet Reconnaissance Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
2 × Reconnaissance Company each	2 × light IFV platoon	6 × BMP-1/2
	2 × infantry platoon	
	1 × heavy tank platoon	3 × T-64/72/80
1 × Reconnaissance Assault Company	3 × light reconnaissance platoon	18 × BRDM-2

Table 29: Soviet Anti-Tank Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
2 × Anti-Tank Gun Battery each	1 × towed anti-tank battery	6 × 100-mm MT-12
	1 × light APC platoon	7 × MT-LB
1 × ATGM Battery	3 × light anti-tank platoon	9 × BRDM-2 ATGM

Table 30: Soviet SSM Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × mobile HQ platoon	
2 × SSM Battery each	1 × mobile missile artillery section	2 × FROG-7 or SS-21

RULE 1: GROUND UNITS

A ground *unit* is a combat or logistics formation of infantry, armor, transports, AAA, SAM launchers, or radars. This section discusses them in general.[85]

A. Type and Size: A ground unit will have a type, for example, “heavy tank” or “ZSU-23-4”, and a size, which will be one of squad, section, platoon, or battery. Typically, a squad has five to ten soldiers and/or one vehicle, gun, or launcher; a section has ten to twenty soldiers and/or two vehicles, guns, or launchers; and a platoon or battery has thirty to fifty soldiers and/or three to six vehicles, guns, or launchers. The term “platoon” is more commonly used for infantry and tank units and “battery” for artillery and air-defense units, but within the game they are equivalent.

Companies, battalions, and other larger units above platoons or batteries are represented by multiple units. For example, an infantry company might be represented by three infantry platoon units.

All ground combat and transport platoons are also available as sections and squads and all AAA platoons and batteries are also available as sections. A section or squad is identical in every way to a platoon or battery that has taken 1D or 2D damage, respectively, except that no VPs are awarded for this notional damage and non-AAA squads and sections may not engage in barrage fire.[86]

In the game, ground units are uniformly referred to as squads, sections, platoons, and batteries. However, other terms are used for these units even within English-speaking armies. For example, in many Commonwealth armies a section is equivalent to a squad, a troop is equivalent to a platoon, and a squadron is equivalent to a company. In the US Armored Cavalry, a troop is equivalent to a company and a squadron is equivalent to a battalion.

B. Basic Properties: The type and size of a ground unit determines its basic properties: damage resilience, defense strength, target class, mobility, and sighting range.

The damage resilience is the amount of damage a ground unit can suffer before being killed. It is determined by the size of the unit: a squad has a damage resilience of 1D; a section has a damage resilience of 2D; and a platoon or battery has a damage resilience of 3D.[87]

The numerical defense strength is used to calculate the odds for air-to-ground attacks and for ground-to-ground combat.

The target class is used to determine the appropriate attack strength of any weapons used in air-to-ground attacks. The class can be a hard, open, or soft according to the degree to which the unit enjoys armored protection. Many weapons have different attack strengths against targets of different classes. A hard target has armored protection from both direct (front, side, and rear) and top attacks. Hard targets are further described as light, medium, or heavy according to their degree of protection, and have corresponding defense strengths.[88] An open target has armored protection from direct attacks but not from top attacks. An open target is normally treated as a hard target, but against napalm, fire, FAE, and air-burst HE bombs, it is considered to be a soft target.[89] A soft target does not have armored protection their personnel and/or weapons.[90]

C. Representation: The briefing documents show the graphical representation of ground units as counters.[91] The size of a unit is indicated by one, two, or three dots (for squads, sections, or platoons) or one vertical bar (for batteries) in an upper position. Hard targets are indicated by an underlined defense strength and open targets by an underline defense strength followed by an exclamation point.

Advanced Rule

D. Mobility: The mobility of a ground unit restricts the hexes it can occupy according to the terrain in and around those hexes. Infantry units, armored units (whether tracked or wheeled), and tracked unarmored units have unrestricted (U) mobility. Trucks and other wheeled unarmored units have good (G) or poor (P) mobility. All other units have towed (T) mobility.[92]

A unit with unrestricted mobility may occupy: any urban hex; any clear hex; or any forest hex. A unit with good mobility may occupy: any urban hex; any clear hex; or any forest hex with a road or trail. A unit with poor mobility may occupy: any urban hex; any clear or forest hex with a road, trail, or runway; or any clear hex adjacent to an urban hex or another clear hex with a road, trail, or runway. A unit with towed mobility may occupy hexes according to the mobility of its transport.[93]

When necessary, a scenario must specify whether wheeled unarmored units have good or poor mobility, according to the terrain and the capabilities of the vehicles represented, and assign transport units to towed units.[94] A scenario may modify the mobility classes of ground units and the associated restrictions. [95]

E. Composite Platoons: At the start of the game, an infantry, infantry weapons, or infantry HQ platoon may attach one or two infantry squads to form a composite platoon. The attached squads are commonly infantry SAM or FAC squads. Once formed, a composite platoon may not be split into its constituent units.[96]

An attached squad is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. Instead, it is considered to form part of the composite platoon. An attached squad retains its normal capabilities (for example, an attached SAM squad may engage aircraft and an attached FAC squad may sight and mark targets).

A composite platoon has the same damage resilience, defense strength, target class, mobility, and sighting range as its constituent platoon. If a composite platoon is sighted, no information is given on any attached squads. If a composite platoon is identified, any attached squads are also identified (for example, “infantry platoon with an attached FAC squad”). If a composite platoon is suppressed, each attached squad is suppressed too. If a composite platoon suffers damage, then for each D of damage, roll one die for each attached squad and on a 3− the attached squad is killed. If a composite platoon is killed, each attached squad is killed too. VPs are awarded for damage to each of the constituent units.[97]

F. Transport: An infantry or towed unit may be transported by a specific associated transporting IFV, APC, truck, or tracked-carrier unit of at least the same size. When necessary, a scenario will establish the association between units and their specific transports. [98]

A transported unit is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. It may not attack other ground units or aircraft, may not launch SAMs, may not use radar, and may not act as a FAC.

If a transporting unit is sighted, no information is given on the transported unit. If a transporting unit is identified, no information is given on a transported infantry unit, but any transported towed unit is also identified (for example, "truck platoon towing an artillery battery" or "truck platoon towing a ZPU-4 battery"). If a transporting unit suffers damage, then its transported unit suffers the same damage. VPs are awarded for damage to both the transporting and transported unit.

A unit may start the game either transported or not. If not transported, it may be placed according to either its mobility or that of its associated transport.[99]

An infantry unit in the same hex as its associated transport may declare that it is mounting the transport at the start of a Ground Unit Interaction Phase. Both units must not be suppressed, and the transport must not have suffered more damage than the infantry unit. At the end of the Ground Unit Interaction Phase five game turns later, the unit is considered to be transported, and its counter is removed from the map.

An infantry unit being transported may declare that it is dismounting at the start of a Ground Unit Interaction Phase. The transport unit must not be suppressed. At the end of the Ground Unit Interaction Phase two game turns later, the unit is considered to be no longer transported, and its counter is placed on the map in the same hex as its transport. If placing the counter would violate stacking requirements, the unit must abort the dismounting process.

During these processes, the infantry unit may not perform any other action and the transport may not move and may not carry out ground attacks, but may perform barrage fire if it has this capability. The infantry unit may abort either of these process at the start of any Ground Unit Interaction Phase. If either are suppressed during the process, the unit must abort the process in the next Ground Unit Interaction Phase.[100]

A towed unit may not mount or dismount during the game.

Table 31: Soviet Motor-Rifle Division (1980s)

Element	Game Ground Unit	Notes
1 × Divisional HQ	1 × armored HQ platoon 6 × SA-7/14/16 squad 6 × light truck squad	
1 × Reconnaissance Battalion		
1 × Anti-Tank Battalion		
1 × SSM Battalion		
1 × BMP Motor-Rifle Regiment		
2 × BTR Motor-Rifle Regiment		
1 × Tank Regiment		
1 × Artillery Regiment		
1 × SAM Regiment		

NOTES

1. **Robin Olds.** The full quotation contains a second insight: "You're not going to win a war by shooting down a damn MiG, you're going to win a war by bombing the bejesus out of whatever you were sent to bomb. Of course, we weren't there to win it anyway, so it was all a waste of time." (Sherwood, John. Fast Movers: Jet Pilots and the Vietnam Experience, chapter 1)

2. **Ground Combat and Transport Units.** *Air Strike* and *Eagles of the Gulf* provide eleven types of ground combat and transport unit: light, medium, and heavy armor, infantry, infantry FAC, infantry SAM, towed artillery, infantry HQ, armored HQ, truck, and light truck. We provide many more. Does this change outcomes? No. Effectively, all ground units are basically a protection class, defense strength, and sighting range, and the new units map to the existing ones. Does this change the experience? I think it does; the wider range of units enhances the gaming experience.

3. **Infantry Weapons Platoon.** An infantry weapons platoon is not strictly necessary, but it adds to the look-and-feel, and we might consider giving them longer range in ground-to-ground combat. Having a single type is probably more appropriate than having separate types for grenade launchers, recoilless rifles, ATGMs, and machine guns. One might think that the .50 cal or DShK infantry AAA unit could serve as a machine-gun weapons platoon, but I would imagine most machine-gun weapons platoons have low tripods for ground combat rather than high ones for AAA.

4. **Heavy Tanks.** Regarding the protection class:

– The WRG 1950–1985 rules give class IX or X (i.e., heavy) protection to the: Chieftain, Conqueror, EPC (Leclerc), IS-3, Leopard 2, Merkava 1, Shir 2 (Challenger 1), Stridsvagn 103, T-10, T-64, T-72, and T-80.
– Al-Kalid/VT-1A and Type 99: These are based on the Type 90-II prototype, which in turn is based on the T-72 (heavy).
– Ariete: This weighs 54 tonnes, only slightly less than the M1A1 (heavy) which also has a crew of four, and therefore presumably has similar protection.
– Arjun: The weights 59–68 tonnes and has crew of four, similar to the M1A1 (heavy), and so presumably has similar protection.
– Challenger 2: This is derived from the Challenger 1 (heavy).
– IS-2: This has a similar weight to the IS-3 (heavy) and a similar armament (122 mm D-25), so presumably similar protection.
– K1: This is derived from the M1 (heavy).
– K2: This presumably has at least the same protection as the K1 (heavy).
– M-84, PT-91, and T-90: These are derived from the T-72 (heavy).
– Merkava 2/3/4: These are derived from the Merkava 1 (heavy).
– T-14: This presumably has at least the same protection as the T-90 (heavy).
– T-84: This is derived from the T-80 (heavy).
– Type 10: This weighs 44 tonnes in its standard configuration and has a crew of three, and so presumably has protection at least equal to earlier tanks with similar weights such as the T-72 (heavy).
– Type 90: The has a crew of three and weights 50 tonnes, and so presumably has protection at least the equal of the lighter T-72 (heavy).
– Type 99: The has a crew of three and weights 51–55 tonnes, and so presumably has protection at least the equal of the lighter T-72 (heavy).

5. **Medium Tanks.** Regarding the protection class:

– The WRG 1950–1985 rules give class IV to IIX (i.e., medium) protection to the: AMX-30, ASU-85, Centurion, Comet, Cromwell, ISU-122, ISU-152, Kanonen JPz, Leopard 1, M103, M4 Sherman, M46, M47, M48, M48, M60, Panzer 68, SU-100, T-34, T-54, T-55, T-62, TAM, Type 61, Type 74, and Vijayanta.
– M26: The M46 (medium) is largely an M26 with an improved engine.
– M-50 and M-51 Sherman: These are based on the M4 (medium) and have improved armament and engines.
– Panzer 61: This is essentially an early version of the Panzer 68 (medium).
– Type 59/69/79: The Type 59 was Chinese version of the T-54A (medium). The Type 69 and 79 had improved equipment, some of which was derived from captured T-62s, but the same level of protection.
– Type 85/88/96: These have new hulls and turrets. It's not clear whether their level of protection is significantly improved over the Type 59/69/79 (medium).
– Type 15: Wikipedia gives little information on the armor, but the weight is 36 tonnes (with the armor package), and that seems to correspond more to a medium tank.
– Vickers MBT: This is similar to the Vijayanta (medium).

6. **Light Tanks.** Regarding the protection class:

– The WRG 1950–1985 rules give class II and III (i.e., light) protection to the: AMX 10RC, AMX 13, Commando, Cougar, Fox, SpPz Luchs, M8, M24, M41, M114A2, M551 Panhard AML, Panhard EBR, PT-76, Saracen, Scimitar, and Scorpion
– Stryker MGS: With additional ceramic armor, this has protection against 14.5 mm but not heavier rounds, which I believe counts as III.
– Strv 74: This has a similar weight to the M41 (light).
– Type 62: Wikipedia give the hull armor as being similar to the M41 (light).
– Type 63: Wikipedia give the hull armor as being only 10 mm, which corresponds to light.
– Type 05: Based on Type 05 IFV (light).
– Type 08: Based on Type 08 IFV (light).

7. **Open Tanks.** These are uncommon, but we need an open target class for M3 half-track APCs, so it is easy to apply it here too. Regarding the protection class:

– The WRG 1950–1985 rules give class I (i.e., open) protection to the: ASU-57 and SU-76.

8. **Anti-Tank Units.** Anti-tank units are mainly chrome. However, in ground combat, we might give them more effectiveness against vehicular or towed targets and less against infantry targets.

9. **Medium Anti-Tank Vehicles.** Regarding the protection class:

– The WRG 1950–1985 rules give class IV (i.e., medium) protection to the: Raketen JPz.
– Jaguar: This is Raketen JPz (medium) upgraded with HOT missiles.

10. **Light Anti-Tank Vehicles.** Regarding the protection class:

– The WRG 1950–1985 rules give class II or III (i.e., light) protection to the: BRDM and Striker.
– Fennek MRAT: Based on the Fennek (light).
– FV438: This is derived from the FV432 APC (light).
– M150 and M901: These are derived from the M113 APC (light).
– M1134 Stryker ATGM: These are derived from the Stryker APC (light).
– VAB HOT: This is derived from the VAB APC (light).
– YP-408 PRAT: This is derived from the YPR-408 APC (light).
– YPR-765 PRAT: This is derived from the YPR-765 APC (light).

11. **Light Reconnaissance Vehicles.** Regarding the protection class:

– The WRG 1950–1985 rules give class II or III (i.e., light) protection to the: BRDM-2, Ferret, Lynx (M113½ Lynx), and M114.
– LAV II Coyote: Wikipedia states that it is protected against small-arms rounds, which corresponds to light.
– Fennek: Wikipedia states that it is protected against small-arms rounds, which corresponds to light.

12. **Light Reconnaissance Vehicles.** Regarding the protection class:

– M20: Open version of the M8 (light) with no turret and armed only with a .50-cal machine gun.

13. **IFV and APC Units.** And again we might distinguish APCs and IFVs in ground-to-ground combat. Bradley IFVs made mincemeat of Iraqi tank regiments; M113 APCs would not have had the same impact.

14. **Medium IFVs.** Regarding the protection class:

– The WRG 1950–1985 rules give class IV (i.e., medium) protection to the: Marder.
– BMP-3: Wikipedia says the BMP-3 has frontal protection from 30 mm rounds at 200 m.
– Boxer: Wikipedia states that with the baseline armor it is protected against medium-caliber rounds. Additional armor is also available.
– Strf 9040/CV90: Wikipedia states that the CV90 has frontal protection against 30 mm APFSDS rounds. This is considerably more than typical light AFVs.
– LAV III (add-on-armor): Wikipedia says the basic version only has protection against small-arms rounds, but an additional armor kit developed in 2009 gives it protection against 30 mm rounds.
– M2A2/M2A3/M2A4: Wikipedia states their these M2A2 also has protection against 30 mm APDS rounds and RPGs. The M2A3 and M2A4 have visually similar armor.
– Puma: Wikipedia says the frontal arc has protection against medium-caliber rounds and shaped-charge projectiles. With additional armor, the sides gain similar protection.

15. **Light IFVs.** Regarding the protection class:

– The WRG 1950–1985 rules give class II or III (i.e., light) protection to the: AMX-10P, BMD, BMP-1, Commando, Dutch AIFV (YPR-765), FV432 (FV432 RARDEN), HS.30 (SPz Lang), M113 (M113 FSC), Spz Kurz, UK MICV (Warrior), and XM723 (M2/M3 Bradley).
– BMP-2: Wikipedia says the BMP-2 has similar protection to the BMP-1.
– LAV-25: Wikipedia says this only has protection against small-arms rounds.
– LAV III: Wikipedia says the basic version only has protection against small-arms rounds, but an additional armor kit developed in 2009 gives it protection against 30 mm rounds.
– Pbv 301: Wikipedia states the armor ranges from 8 to 50 mm, with upper front armor being 34 mm (20 mm at 55 degree). This seems to be light.
– Piranha IFV: Presumably similar protection to LAVs.
– Strkyer ICVD: This is derived from the Stryker (light).
– Type 86: Wikipedia states this is reverse-engineered from the BMP-1.
– Type 92: Wikipedia states that it is protected against small-arms rounds and the frontal armor can stop 12.7 mm rounds at 100 meters and 25 mm rounds at 1000 meters. This seems to correspond to light armor. The weight is also 21 tonnes, which is considerable less than most medium IFVs.
– Type 05: Wikipedia states that it has protection against small-arms rounds and some protection against 12.7 mm rounds. This corresponds to light armor.
– Type 08: Wikipedia states that it is protected against small-arms rounds and the frontal armor can stop 12.7 mm rounds at 200 meter. This corresponds to light armor.
– VCBI: Wikipedia says this only has protection against 14.5 mm rounds.

There are many variants of the Piranha, and the in-service date is a guess.

16. **Heavy APC.** Regarding the protection class:

– Achzarit: Wikipedia states that this based on the T-54/T-55 (medium), but has additional composite armor. Its weight at 44 tonnes is much heaver than the T-55 at 36 tonnes, despite the lack of turret, which suggests the additional armor is quite significant. Therefore, we consider it to be heavy.
– Namer: This is based on the Merkava IV (heavy).
– Nakpadon: This is an improved version of the Nagmashot and Nagmachon. Wikipedia states that its weight is around 55 tonnes, which strongly suggests its armor qualifies as heavy.
– Puma: The photos in Wikipedia show additional armor. Its weight at 50 tonnes is almost identical to the Centurion, despite the lack of turret, which suggests the additional armor is quite significant.

The in-service dates for the Nakpadon (1998) and Puma (1993) are guesses.

17. **Medium APCs.** Regarding the protection class:

– Boxer: Wikipedia states that with the baseline armor it is protected against medium-caliber rounds. Additional armor is also available.
– Nagmachon: Based on the Centurion (medium).
– Nagmachon: Based on the Centurion (medium).

The in-service dates for the Nagmashot (1990) and Nagmachon (2000) are guesses.

18. **Light APCs.** Regarding the protection class:

– The WRG 1950–1985 rules give class II or III (i.e., light) protection to the: BTR-50PK, BTR-60PB, BTR-152K, Commando, Grizzly, FV432, LVTP-7, M113, M59, Saracen, Spartan, VAB, and YPR-408.
– Bronco/Warthog: Wikipedia says it has protection against 7.62 mm rounds.
– BVS10/Viking: Wikipedia says it has protection against small-arms fire and can have additional armor to protect against 14.5 mm rounds.
– BTR-70: Wikipedia states the frontal armor is only 9 mm.
– BTR-80: Wikipedia states that this (and previous models) only have protection from small-arms fire.
– LAV II Bison: Wikipedia states that the Coyote is protected against small-arms rounds, and the Bison seems to be similar.
– LVTP-5: Wikipedia states that the maximum armor is 16 mm.
– M75: Wikipedia states the maximum armor is 38 mm. This is more than the thickness of some light tanks, but it is almost vertical.
– OT-64 SKOT: Wikipedia states the maximum armor is 13 mm.
– OT-810: Wikipedia states the maximum armor is 14.5 mm.
– Piranha APC: Presumably similar protection to LAVs.
– M1126 Stryker: Wikipedia states it has some protection against 14.5 mm rounds from the front and all-round protection against small-arms fire.
– Saxon: Wikipedia states that it is protected against small-arms rounds.
– TPz Fuchs: Wikipedia states that it is protected against small-arms rounds.
– Type 63: Wikipedia states that it is protected against small-arms rounds.
– Type 77: Wikipedia states that it is based on the PT-76 (light), and its weight of 15-5 tonnes matches other light APCs.
– Type 89: Wikipedia states that it is protected against small-arms rounds.
– Type 08: Based on Type 08 IFV (light).

There are many variants of the Piranha, and the in-service date is a guess.

19. **Open IFVs.** Regarding the protection class:

– The WRG 1950–1985 rules give class I (i.e., open) protection to the: AT-P, BTR-40, BTR-50P, BTR-60P, BTR-152A, and Half-tracks (M3).
– SKP/VKP/SKPF/VKPF: Wikipedia states the armor is 8 to 20 mm, which is light.

20. **HQ Units.** I wondered if HQ platoons were also used at the company level, but looking at the *Air Strike* scenarios they are included at a ratio of about one HQ platoons to nine normal platoons, so they seem to be battalion HQs.

21. **Armored Mortars.** Regarding the protection class:

– FV432: based on FV432 (light).
– M106/M125/M1064: based on M113 (light).
– M1129 Stryker MCV: based on Stryker (light).

We might consider these as open.

22. **Light Armored Artillery.** Regarding the protection class:

– The WRG 1950–1985 rules give class II (i.e., light) protection to the: Abbot, AMC-155GCT (AMX-30 AuF1), L33, M52, M108, M109, SP.73 (2S3), SP.74 (2S1), and VK.155 (BKan 1).
– 2S19 Msta-S: Wikipedia states this all-round protection from 15 mm of steel armor. This corresponds to light armor.
– AS-90: Wikipedia states this has up to 17 mm of steel armor. This corresponds to light armor.
– Panzerhaubitze 2000: Wikipedia states this all-round protection from 14.5 mm of steel armor. This corresponds to light armor.

23. **Open Armored Artillery.** Regarding the protection class:

– The WRG 1950–1985 rules give class I (i.e., open) protection to the: AMX-105A, M7, and M44.
– The M-50 has frontal and side armor, like the M7 and unlike for example the M107 and M108.
– The M40 also has frontal and side armor. The rear door is probably armored too, but is open when the gun is used.

24. **Tracked Artillery.** Regarding the protection class:

– The WRG 1950–1985 rules give class I (i.e., open) protection to the AMX.155F3 (AMX-13 F3), M107, and M110. However, the only crew member protected seems to be the driver. The other crew members travel and operate the gun without protection.
– Similar considerations apply to the: 2S4, 2S7 (although some of the crew travel under armor), M40, M41, and Mk F3. In all of these, the gun mount is open.

25. **Mobile Artillery.** Regarding the protection class:

– CAESAR: Some models have lightly-armored cabins, but the truck as a whole does not seem to be protected.
– Archer: The crew and engine compartments is protected against small-arms rounds, but the gun does not seem to be protected. Unlike the CAESAR, the gun is operated by the crew from the protected cabin. This is close to reaching the level of protection of light armored artillery.

26. **Light Armored Artillery.** Regarding the protection class:

– M270: I can find no definite information on the armor protection.
– BM-1: According to Wikipedia, the T-72 hull was chosen to give adequate protection close to the front lines. However, it is not clear whether the rocket launcher has adequate protection.

27. **Mobile Rocket Artillery.** Regarding the protection class:

– M142 HIMARS: The cabin has some protection, but the rest of the truck seems to be unprotected.
– BM-30: The cabin appears to have some protection, but the rest of the truck seems to be unprotected.
– Earlier Soviet vehicles appear to be completely unarmored.

28. **Mobile Rocket Artillery.** Regarding the protection class:

– M752 Lance: This seems to be based on the unarmored M548.
– AMX-30 Pluton: The vehicle is based on an AMX-30 hull and seems to be armored, but the missile is exposed.

Tracked missile artillery is mainly included to serve as a target.

29. **Missile Artillery Batteries.**

Regarding the protection class, none of these have complete protection against small arms.

Missile artillery is mainly included to serve as a target.

30. **Tracked Carrier Vehicles IFVs.** All the listed tracked carriers are stated as being unarmored by Wikipedia.

31. **AAA Values.** This long note summarizes how I have determined the AAA values for AAA units. VP values are discussed separately.

I compare AAA values for AAA units from *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat* to the actual characteristics of the guns they represent. My aim is to empirically understand the underlying model sufficiently to identify anomalous values and to apply it to other guns. That said, I do comment on the model.

AAA Values for Batteries. I start by attempting to understand the AAA values for batteries.

The upper part of the following table shows the original data for gun batteries from *Air Strike* (AS), *Eagles of the Gulf* (EOTG), and *The Speed of Heat* (TSOH). I have added columns with the caliber, burst rate of fire per mount, and muzzle velocity.

Type	Maximum Altitude	Ranges	Hit Rolls	Damage	FCR	Caliber (mm)	Rate of fire (RPM)	Muzzle Velocity (m/s)	Source
Original									
M2/DShK-38	5	2/3/4	3/2/1	1	—	12.7	500	890	AS
M55	5	2/3/4	5/4/3	2	—	12.7	2000	890	EOTG
ZPU-1	6	2/3/5	3/2/1	1	—	14.5	600	1000	EOTG/TSOH
ZPU-2	6	2/3/5	4/3/2	1	—	14.5	1200	1000	AS
ZPU-4	6	2/3/5	5/4/3	2	—	14.5	2400	1000	AS
ZPU-4	6	2/3/5	4/3/2	2	—	14.5	2400	1000	TSOH
TCM-20	8	2/4/6	4/4/3	2	—	20	1450	860	EOTG
Single Rh-202	8	2/4/6	3/2/1	2	—	20	1030	1044	AS
Twin Rh-202	8	2/4/6	5/4/3	2	—	20	2060	1044	AS
M167	8	2/4/6	6/5/4	3	R	20	3000	1030	AS
ZU-23	9	2/5/8	4/3/2	2	—	23	2000	980	AS/TSOH
Gemag	10	3/6/9	6/5/4	4	R	25	1800	1040	AS
Oerlikon GDF	12	4/8/10	5/4/3	4	—	35	1100	1175	AS
61-K (M-38)	12	4/8/11	3/2/1	4	—	37	165	880	AS/TSOH
Bofors L/60	15	4/8/12	2/2/1	4	—	40	135	865	EOTG
Bofors L/70	15	4/8/12	3/3/2	4	—	40	330	1000	AS
BOFI	15	4/8/12	4/4/3	4	W	40	330	1000	AS
S-60	18	5/10/15	2/2/1	5	—	57	110	1000	AS/TSOH
KS-12	27	6/12/18	2/1/0	6	—	85	20	790	AS/TSOH
KS-19	39	7/14/21	1/1/0	6	—	100	15	900	AS
Modified									
ZPU-4			5/4/3		— for rate of fire				
Single Rh-202			4/3/2		— for rate of fire				
Twin Rh-202			3		— for rate of fire				
ZU-23			5/4/3	3	— for rate of fire				
Oerlikon GDF			4/3/2		— for rate of fire				
BOFI			5/5/3		— for FCR-W				

Qualitatively, altitude and range increase with caliber, hit rolls increase with rate of fire, and damage rating increases with both caliber and rate of fire. This is as expected.

Quantitatively:

– Approximately doubling the rate of fire increases the hit rolls by one, with typical values of 3/2/1 around 500 rounds per minute, 4/3/2 around 1000, and 5/3/2 around 2000 and above.
– The base damage rating is 1 for 12.7–14.5 mm, 2 for 20–23 mm, 3 for 25–30 mm, 4 for 35–40 mm, 5 for 57 mm, and 6 for 85–100 mm.
– Damage rating by one is increased by one if the rate of fire is 1800 rounds per minute or more.

Most of the original values can be used as they are, but a few of the hit rolls need adjusting:

– ZPU-4 and ZU-23:
There are six AAA batteries with rates of fire per mount of about 2000 rounds per minute or more: the M55, ZPU-4, twin Rh-202, M167, ZU-23, and Gemag. After accounting for the +1 modifier for the FCR-R of the M167 and Gemag, all have hit rolls of 5/4/3 except for the ZPU-4 from *The Speed of Heat* (although the one from *Air Strike* does have 5/4/3) and the ZU-23.
Furthermore, all of these except for the ZU-23 (both *Air Strike* and *The Speed of Heat* versions) have damage ratings increased by one above their baseline values.
It seems that either there is some reason to reduce the effectiveness of the ZPU-4 and ZU-23 compared to these other guns, or perhaps they have inconsistent values rolls.
The ZU-23 is undeniably basic, and one might think that would be reason to reduce its effectiveness. However, the TCM-20 is basic too and has both a lower rate of fire than the ZU-23 (1450 compared to 2000) and a smaller caliber (20 mm compared to 23 mm), yet its hit rolls are higher than the ZU-23 in *The Speed of Heat* and its damage rating is the same.
The best solution would seem to be to give these guns the values we would expect from comparing them to other guns with similar rate of fire and caliber. Therefore, I give both the ZPU-4 and ZU-23 hit rolls of 5/4/3 and the ZU-23 a damage rating of 3.
(I also considered the idea that increasing the rate of fire increases *either* the hit rolls *or* the damage rating. However, while this might explain the ZPU-4 from *The Speed of Heat* and the twin Rh-202, it does not correctly predict the values of the M167 or Gemag.)
– Single RH-202
The single Rh-202 has unexpectedly low original hit rolls of 3/2/1 for its rate of fire. The values of 4/3/2 for the ZPU-2 and TCM-20, which have a similar rate of fire, are probably more appropriate.
– Twin Rh-202
The twin has an unexpectedly low original damage rating of 2 for its rate of fire. A value of 3, increased by one from the baseline for the caliber, is probably more appropriate.
– Oerlikon GDF
The Oerlikon GDF has unexpectedly high original hit rolls of 5/4/3 for its rate of fire. The values of 4/3/2 for the ZPU-2, which has a similar rate of fire, are probably more appropriate.
– BOFI
The BOFI has unexpectedly low original hit rolls of 4/4/3. Values of 5/5/3, obtained by applying a +2 modifier for FCR-W to the Bofors L/70, are probably more appropriate. (Note that the "BOFI" in *Air Strike* corresponds to what I call the "BOFI-R" below.)

I adopt these modified values.

Also, I note the following:

– FCR-R
The M167 and Gemag have hit rolls of 6/5/4, and the twin Rh-202 has 5/4/3. Otherwise, they are similar, except that the M167 and Gemag have FCR-R, and the Rh-202 does not (and the M167 has a moderate advantage in rate of fire). I consider the FCR-R of the M167 and Gemag to be largely responsible for increasing their hit rolls by +1 compared to the Rh-202.
– Bofors
The hit rolls of 2/2/1 for the Bofors L/60, 3/2/1 for the 61-K (M-38), and 3/3/2 for the Bofors L/70 seem to reflect the increase in rate of fire.
The L/60 and L/70 have the same maximum altitude of 16 and maximum range of 12, despite the L/60 having lower muzzle velocity. However, I am not especially motivated to attempt to change this, since the Bofors is primarily a weapon for point-defense against aircraft up to about level 10.
The real BOFI and BOFI-R also have proximity fuzes, and I consider these will increase their hit rolls by +1.

AAA Values for Sections. I now turn to sections. Again, my aim is to empirically understand the underlying model sufficiently to identify anomalous values and to apply it to other guns.

The upper part of the following table shows the original data for gun sections from *Air Strike* (AS), *Eagles of the Gulf* (EOTG), and *The Speed of Heat* (TSOH). Again, I have added columns with the caliber, burst rate of fire per mount, and muzzle velocity.

Type	Maximum Altitude	Ranges	Hit Rolls	Damage	FCR	Caliber (mm)	Rate of fire (RPM)	Muzzle Velocity (m/s)	Source
				Original					
M16	5	2/3/4	4/3/2	2	—	12.7	2000	890	EOTG
Mobile ZPU-4	6	2/3/5	3/3/2	2	—	14.5	2400	1000	AS
M163	8	2/4/6	5/4/3	3	R	20	3000	1030	AS
M3 TCM-20	8	2/4/6	3/3/2	2	—	20	1450	860	EOTG
Panhard	8	2/4/6	2/2/1	2	—	20	1450	860	AS
Mobile ZU-23	9	2/5/8	2/2/1	2	—	23	2000	980	AS
Nile 23	9	2/5/8	3/3/2	2	—	23	2000	980	EOTG
ZSU-23-4	9	2/5/8	5/4/3	3	W	23	4000	980	AS
Blazer	10	3/6/9	5/4/3	4	W	25	1800	1040	AS
M53/59	11	4/7/9	4/3/2	3	—	30	1000	1000	AS
AMX-30	11	4/7/9	5/4/3	3	W	30	1200	970	AS
ZSU-30-2	11	4/7/9	5/4/3	4	W	30	5000	960	AS
Gepard	12	4/8/10	5/4/3	4	W	35	1100	1175	AS
ZSU-57-2	18	5/10/15	2/1/1	5	—	57	240	1000	EOTG
				Modified					
Mobile ZPU-4			4/3/2		— to match the ZPU-4				
Panhard			3/3/2		— to match the M3 TCM-20				
Mobile ZU-23			4/3/2	3	— to match the ZU-23				
Nile 23			4/3/2	3	— to match the ZU-23				
M53/59			3/2/1		— to match the AMX-30				

Comparing similar weapons, batteries and sections have the same maximum altitudes, ranges, and damage ratings. However, batteries (with four to six mounts) typically have a +1 advantage on their hit rolls compared to sections (with two mounts). In a very few cases, the difference is 0, and in a very few others, it is +2. A difference of 1 in the hit rolls is commensurate with the approximate doubling of the rate of fire from a section to a battery.

To determine the appropriate AAA for sections, the following recipe seems to work in most cases:

– Take the value for an equivalent battery without FCR and apply an adjustment of −1 to the hit rolls to account for the reduced rate of fire in the section. (Only do this if the rate of fire is reduced; see the comment on the ZSU-23-4 and ZSU-57-2 below.)

– Apply an adjustment of +1 to the hit rolls for FCR-R or similar.

– Apply an adjustment of +2 to the hit rolls for FCR-W.

– The preceding steps would then predict hit rolls of 6/5/4 for the Blazer and 7/6/5 for the ZSU-23-4 and ZSU-30-2. To conform to the original values, I limit the hit rolls to 5/4/3. It would appear that in the underlying model, beyond a certain rate of fire, other factors take over to limit the probability of a hit.

Again, most of the original data can be used as they are, but a few of the values are worth commenting upon or need adjusting.

– Agreements

The original hit rolls and the predicted hit rolls are in agreement for the M16 (from the M55 battery), M3 TCM-20 (from the TCM-20 battery), M163 (from the M167 battery), the Blazer (from the Gemag), and the Gepard (from the Oerlikon GDF).

– Mobile ZPU-4

The original hit rolls for the mobile ZPU-4 section are 3/3/2, but 4/3/2 is predicted from the values of 5/4/3 for the battery and is in better agreement with the values for the M16.

– Panhard M3 DCA

The Panhard M3 DCA has unexpectedly low original hit rolls of 2/2/1 compared to the TCM-20 battery and the M3 TCM-20 section, despite having very similar guns. Values of 3/3/2 are probably more appropriate.

– Mobile ZU-23 and Nile 23

The original hit rolls for the mobile ZU-23 and Nile 23 section are 2/2/1 and 3/3/2, despite them being quite similar. The predicted values of 4/3/2 are probably more appropriate and are in better agreement with the values for the M16. Also, as with the ZU-23 battery, a damage rating of 3 is probably more appropriate given their high rate of fire.

– M53/59

The original hit rolls for the M53/59 section are 4/3/2. The predicted values of 3/2/1, derived from the modified hit rolls of 4/3/2 for the Oerlikon GDF battery and also from the AMX-30SA after discounting the FCR-W, are probably more appropriate.

– ZSU-57-2

The ZSU-57-2 section has the same hit rolls as the S-60 battery. This is probably appropriate, as the ZSU-57-2 is a twin mount and this largely compensates for the reduced rate of fire of the section. The same applies to the quad ZSU-23-4 compared to the twin ZU-23.

I adopt these modified values.

Also, I note:

– Panhard M3 DCA

The Panhard M3 DCA has an FCR-R capability, which is discussed below but not considered here.

– AMX-30 DCA

The real AMX-30 DCA does not have an FCR. The AMX-30 DCA in the game seems to correspond to the similar AMX-30SA, which does.

Summary. I have developed an empirical understanding of the model for AAA values. Consideration of outliers suggests:

– Increasing in the hit rolls of the mounted ZPU-4, single Rh-202, Panhard M3 VDA, ZU-23 (and derivatives), and BOFI.

– Decreasing the hit rolls of the Oerlikon GDF.

– Increasing the damage rating of the twin Rh-202 and the ZU-23 guns (and derivatives).

I have adopted these changes.

32. **M2 Platoon.** The AAA values are from *Air Strike*.

33. **DShK-38 Platoon.** The AAA values are from *Air Strike*.

34. **M16 Section.** The AAA values are from *Eagles of the Gulf*.

35. **M55 Battery.** The AAA values are from *Eagles of the Gulf*.

36. **ZPU-1 Battery.** The AAA values are from *Eagles of the Gulf* and *The Speed of Heat*.

37. **ZPU-2 Battery.** The AAA values are from *Air Strike*.

38. **ZPU-4 Battery.** The AAA values are from *Air Strike* rather than *The Speed of Heat*, for the reasons described in the extended note above.

39. **Mobile ZPU-4 Section.** AAA values are from *Air Strike*, except that I have adopted hit rolls of 4/3/2 rather than 3/3/2 as described in the extended note above.

40. **Mobile ZPU-1 and ZPU-2 Section.** These do not appear in *Air Strike*, *Eagles of the Gulf*, or *The Speed of Heat*, but they are very common. As described in the extended note above, I have produced AAA hit values for the sections from those of the ZPU-1 and ZPU-2 batteries.

41. **M167 Platoon.** The AAA values are from *Air Strike*. I'm not sure when it gained FCR-R or if it gained night IR sights.

42. **M163 Section.** The AAA values are from *Air Strike*. I have the same uncertainties as I do for the M167.

43. **TCM-20 Platoon.** The AAA values are from *Eagles of the Gulf*.

44. **M3 TCM-20 Section.** The AAA values are from *Eagles of the Gulf*.

45. **Rh-202 Battery.** *Air Strike* has single and twin Rh-202 batteries, but all AAA mounts seem to be twin. The AAA values are from *Air Strike*, with the hit values for the single mount changed from 3/2/1 to 4/3/2 and the damage value for the twin mount changed from 2 to 3, as described in the extended note above.

46. **Panhard M3 DCA Section.** The AAA data are from *Air Strike*, but modified as described in the extended note above to give hit rolls of 3/3/2 without FCR. However, the Panhard M3 DCA radar-equipped vehicle seems to be able to search and share speed and range information with the other vehicle in the section, which seems equivalent to FCR-R. Therefore, I have increased the hit rolls by one to give 4/4/3 with FCR-R.

47. **ZU-23 Battery.** The AAA values are from *Air Strike*, but the hit rolls are modified as from 4/3/2 to 5/3/2 and the damage rating from 2 to 3 as described in the extended note above.

48. **Mobile ZU-23 Section.** The AAA values are from *Air Strike* and *The Speed of Heat*, but the hit rolls of 4/3/2 were obtained from those for the ZU-23 battery using the recipe described above, and the damage was modified from 2 to 3 as described in the extended note above.

49. **Sinai 23 Section.** The turret seems to be similar to the LAV-AD turret, but with a pair of 23 mm guns instead of a GAU-12 25 mm gun. The AAA values were taken from the mobile ZU-23 battery. I don't know which model of Stinger was used; I would guess the FIM-92A.

50. **ZSU-23-4 Section.** The AAA values are from *Air Strike*.

51. **LAV-AD Section.** The system seems similar to the Blazer, but without radar and with night IR sights. I've adopted the range and damage rating for the Gemag battery in *Air Strike* with the hit rolls reduced by one for being a section and one again for not having FCR-R giving 4/3/2.

52. **M53/59 Section.** The AAA values are from *Air Strike*, but the hit rolls were reduced from 4/3/2 to 3/2/1 as described in the extended note above. If the VADS is classed as armored, should this be armored too?

53. **AMX-30SA Section.** I adopt the *Air Strike* values given for the AMX-30 DCA. The real DCA does not have radar, but the *Air Strike* one does, matching the SA.

54. **Tunguska Section.** The AAA values are from *Air Strike* for the ZSU-30-2.

55. **Tunguska-M Section.** The AAA values are from *Air Strike* for the ZSU-30-2.

56. **Oerlikon GDF Battery.** The AAA values for the battery are from *Air Strike* with the hit rolls reduced from 5/4/3 to 4/3/2 as described in the extended note above.

57. **Gepard Section.** The AAA values are from *Air Strike*.

58. **61-K Battery.** For the 61-K battery: The AAA values are from *Air Strike* and *The Speed of Heat*.

59. **M-38 Battery.** I can find no sources other than *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat* that refer to the 61-K 37 mm gun as the "M-38". Another designation for the gun is "M-1939", so I could understand "M-39".

60. **Bofors L/60 Battery.** The AAA values are from *Eagles of the Gulf*.

61. **Bofors L/70 Battery.** The AAA values are from *Air Strike*.

62. **Bofors L/70 BOFI and BOFI-R Battery.** The AAA values are adapted from *Air Strike* values for the Bofors L/70, with the hit rolls increased by one for proximity fuzes, one for the BOFI laser rangefinder (effectively FCR-R), and another one for the BOFI-R radar, giving hit rolls of 5/5/4 for BOFI and 6/6/5 for BOFI-R whereas *Air Strike* gives the BOFI-R 5/5/4.

63. **BOFI Service.** I don't know when the BOFI and BOFI-R became available or who else uses them.

64. **S-60 Battery.** The AAA values are from *Air Strike* and *The Speed of Heat*.

65. **ZSU-57-2 Section.** For the ZSU-57-2 section: The AAA values are from *Eagles of the Gulf*.

66. **KS-12 Battery.** The AAA values are from *Air Strike* and *The Speed of Heat*.

67. **KS-19 Battery.** The AAA values are from *Air Strike*.

68. **SON-9.** Wikipedia states this was used with the 57, 85, 100, and 130 mm guns and was widely deployed in the Vietnam War. In *The Speed of Heat* the 57 mm S-60 and 85 mm KS-12 are often combined with a FCR-A, presumably the SON-9. The service history is from Wikipedia.

69. **Super Fledermaus.** Wikipedia states there were two versions: the Feuerleitgerät (FLt Gt) 63 (from 1965) and the Feuerleitgerät 69 (from 1970). The latter had "background clutter suppression," which probably is equivalent to MTI.

Wikipedia also states these the prototype of the Gepard used a Flt Gt 63. Since the FCR-W radar of the Gepard is equivalent to a +2 modifier and has VF frequency, it corresponds to an FCR-C. We assume the Super Fledermaus does too.

70. **SA-7A Squad.** The launcher and SAM values are from *Air Strike* with significant modifications:

– Zaloga gives an in-service date of 1968.

– *Air Strike* gives this a seeker type of I, but that does seem appropriate for an uncooled PbS seeker. Instead, I use a seeker type of E, by analogy with the uncooled seeker in the AIM-9B, and works at about 2 microns.

– *Air Strike* gives a turn rate of HT, but the control capability seems to be the same as in the SA-7B and SA-15. I will adopt BT.

– *Air Strike* gives this a speed of 8. Zaloga gives a lifetime of 8 seconds and an average speed of 1550 km/h. This corresponds to a range of 5.2 km or about 10 hexes.

71. **SA-7B Squad.** The launcher and SAM values are from *Air Strike* with significant modifications:

– *Air Strike* gives this a seeker type of I. That seems about right. The seeker was PbS and cooled electrically, like the AIM-9E which also has a type I seeker.

– *Air Strike* gives a turn rate of BT, which I adopt.

– *Air Strike* gives this a speed of 10. Zaloga states that it has a slightly more powerful rocket motor than the SA-7A, but the mean speed is the same at 1550 km/h. I adopt a speed of 10, like that of the SA-7A.

72. **SA-14 Squad.** The launcher and SAM values are from *Air Strike* with significant modifications:

– The seeker is cryogenically cooled at works at about 4 microns (Zaloga). *Air Strike* gives this a type A seeker, but that does seem appropriate; the first type A seekers in Soviet AAMs did not appear until 1985 in the AA-8C. Also, type A seekers have all-aspect capability Zaloga says that the seeker has similar performance to the RIM-43C, but only limited all-aspect capability. A type M seeker seems to be more appropriate.

– *Air Strike* gives a turn rate of ET, but the control capability seems to be the same as in the SA-7A and SA-7B. I will adopt BT.

– *Air Strike* gives this a speed of 12. Zaloga gives a lifetime of 8 seconds and an average speed of 1440 km/h (as it is slightly heavier than the SA-7). This corresponds to a range of 4.8 km or about 9 hexes.

– The warhead is the same as on the SA-7A/7B.

73. **SA-16/18 Squad.** The launcher and SAM values are from Malcolm Pipes with significant modifications:

– The warhead of the SA-16/18 is 1.17 kg, identical to the 1.17 kg of the SA-7 and similar to the 1.1 kg of the SA-14. Only later Iglas had heavier warheads. Therefore, I have reduced the damage ratings to match the SA-7.

– Carl comments that the warhead explodes any remaining rocket fuel. I increase the damage rating by one if the target is reached in the first half of the FPs.

– I have reduced the lock-on range of the SA-16 to match that of the SA-14.

– I have increased the flare rating of the SA-16 to match that of the SA-14.

– I have given both missiles the same speed and turn rate. Zaloga states that the engagement range increased by about 1 km with respect to the SA-7/14, so I am increasing the speed by 2 to 12. I will give both a turn rate of BT/2, matching Stinger.

74. **FIM-43C Squad.** The launcher and SAM values are from *Air Strike*.

75. **FIM-92A/B/C/D Squad.** The launcher and SAM values for the A model are from *Air Strike*. Those for the B/C/D models are from Malcolm Pipes compendium.

– I have increased the flare susceptibility of the FIM-92A from 2 to 3. It did not seem to have significant protection.

– Wikipedia gives an effective range of 5 miles, which is equivalent to a speed of 15. I have left the speed at 14, as that is close enough.

– Wikipedia gives a peak speed of 1930 mp and a self-destruct time of 17 seconds.

– *Air Strike* has the visibility as 2, but looking at photos I don't see much differ

76. **Blowpipe Squad.** The launcher and SAM values are from *Air Strike*.

77. **Javelin/Starburst Squad.** The launcher and SAM values for Javelin are from *Air Strike*. The launcher and SAM values for Javelin are just those for Javelin but with the OG guidance replaced by LG guidance. I have seen mention of a Javelin LML; did anyone use it or was it just a prototype?

78. **Starstreak Squad.** The launcher and SAM values are from Malcolm Pipes. The real darks do not have proximity fuzes, but the missile in the game can obtain a proximity hit. Perhaps a proximity hit here means a hit with only one die?

79. **Starstreak LML Squad.** The SAM values are from Malcolm Pipes. I have guessed the launcher values are the same as for shoulder-launched version, just that the launcher has three ready missiles. I don't know if the launcher has reloads. Is the launcher really portable or does it have to be transported by a vehicle?

80. **Mistral Squad.** The launcher and SAM values are from *Air Strike*. Is the launcher really portable or does it have to be transported by a vehicle?

81. **RBS 70 Squad.** For the RBS 70: The launcher and SAM values are from *Air Strike*. Is the launcher really portable or does it have to be transported by a vehicle?

82. **Ayn-al-Saqr Squad.** The launcher and SAM values are taken to be equal to those of the SA-7B.

83. **British Army Units.** The generic British Army formations from 1960 and 1985 are from the document "Modern British TOEs" by Richard A. Rinaldi (2002).

84. **Soviet Units.** These are from US Army FM 100-2-3 from 1991.

85. **Rules.** Much of this section should be included in the 3A rules, but it's here for the moment.

86. **Squads and Sections as Damaged Platoons.** So for example sections/squads have 2D/1D damage resilience and +1/+2 modifiers to AAA attacks. Similar modifiers should be applied to ground-to-ground attacks.

87. **Damage Resilience in *Air Strike*.** There is a form of damage resilience in *Air Strike*, in that units whose counters have white text are only able to take 1D of damage before being killed (see the last paragraph of rule 28). Such units include infantry SAMs, armored CCUs, and many vehicle SAMs. TSOH also mentions units being able to take only 2D damage, but it doesn't have any relevant units since it attaches SAM teams to infantry platoons rather than having them as separate (see below) and doesn't have a separate FAC unit. In both cases, the idea seems to have been incompletely implemented in the rules. I'm generalizing and homogenizing the idea. This is important as we need sections for mobile AAA units with two vehicles and for the Oerlikon GDF (as Skyguard can only control two guns and not a full battery). To a large degree, a section is equivalent to a platoon that has taken 1D and a squad is equivalent to a platoon that has taken 2D, so the mechanics of the game are really not impacted.

88. **Hard Targets.** Light, medium, and heavy armor correspond to defense strengths of 4, 6, and 8 in *Air Strike*.

Light armor give protection from 14.5 mm and smaller round. Medium armor gives protection from autocannons and guns up to about 90 mm. Heavy armor gives protection against guns of 100 mm or more. Obviously, this is a gross simplification as each vehicle has its own strengths and weaknesses and its armor varies between its front, side, rear, and top. Nevertheless, this rough classification seems to map to the vehicles depicted in scenarios in *Air Strike* and *Eagles of the Gulf*.

The protection level of tanks and other AFVs are based on the armor classes in the Wargames Research Group 1950–1985 rules: open is class I, light is class II to III, medium is class IV to IIX, and heavy is class IX and X. More modern tanks are guesses, and I'm willing to be corrected.

89. **Open Targets.** The BTR-50P APC and the M3 half-tracked APC are examples of open targets. In the 1956 and 1967 Arab-Israeli Wars where both sides used M3 half-tracks. If an M3 is attacked by an air-burst or napalm bomb, the crew and passengers will suffer more or less as if they were in the open.

90. **Soft Targets.** Examples of soft targets include infantry, towed artillery, unarmored vehicles, and armored vehicles that do not fully protect their personnel and/or weapons such as the M-107 and M-110 tracked artillery vehicles.

Nothing like the M-107 or M-110 is included in *Air Strike*, but again I would assert that it is tracked mainly for mobility. Effectively, these are towed artillery installed in an exposed position on a tracked hull and have almost no protection for most of the crew.

Most tracked SAM vehicles are classified as hard targets in *Air Strike* and *Eagles of the Gulf*, despite many of them having completely exposed missiles. If an air-burst bomb explodes above a SAM-4 launcher, it is not going to shrug it off. I think many SAM vehicles are tracked mainly for *mobility* and perhaps for crew protection, but in terms of a mission-kill on their weapon systems, they are soft targets. I've not changed anything yet, but we need to think about this.

91. **Representation.** The representation is based on NATO symbols, including those for infantry, armor, anti-armor, air defense, guns, missiles, mortar, tube artillery, and rocket artillery. One addition is that a rectangle denotes an unarmored vehicle such as a truck or tracked carrier. A truck or mobile unit has two wheel symbols in the lower position, whereas a tracked carrier has a tracked symbol. The mortar symbol is modified to be simply an open circle, for compactness.

All vehicular units have either an armored or unarmored vehicle symbol, all infantry units have an infantry symbol, and towed units have none of these.

The letter T in the central position indicates a transport capacity (e.g., APCs, IFVs, cargo trucks, and tracked cargo carriers). Words and letters in the lower position indicate variants on a basic type: the letters H, M, L, and O on an armored unit indicate heavy, medium, light, and open armored protection; the word WPN or FAC on an infantry unit indicates a weapons or FAC unit; and the letter H on an infantry SAM unit indicates a heavy version of the launcher (e.g., the LML version of Starstreak).

92. **Mobility.** The inspiration for this rule was this photograph of the Mitla Pass from 1967, which shows trucks destroyed on and just off a road. Also, in the the Falklands War roads could be traversed by Landrovers and AMLs, but off-road terrain was impassable to everything except the CVR(T)s and Bv 202s despite being open.

93. **Mobility Restrictions.** When trucks have good mobility, the rule is: don't put trucks or towed weapons in forests unless there is a road. When trucks have poor mobility, the rule is effectively: if the terrain is difficult, keep your trucks and towed weapons near roads even in clear terrain.

94. **Truck Mobility.** For example, a scenario set in the open desert of southern Iraq might specify that trucks have good mobility, but one set in the Falkland Islands might specify that they have poor mobility. Similarly, a scenario might specify that cargo trucks have poor mobility, but mobile artillery units have good mobility. Finally, a scenario might not assign an explicit transport unit to a towed artillery battery, but might state that it has been transported by helicopter and so can be placed as if it had a transport unit with good mobility.

95. **Modifying Mobility Classes.**

One common modification is to explicitly indicate the hexes occupied by each ground unit; in this case this rule is largely superfluous (although if we allow ground units to move, then the restrictions can still be relevant).

Panhard AMLs are wheeled armored vehicles and so normally have unrestricted mobility. Nevertheless, in the Falklands War they were were restricted to roads and trails, and this corresponds to poor mobility. A scenario could fix this problem by explicitly giving them poor mobility.

96. **Composite Platoons.** This rule comes from TSOH. It doesn't have counters for infantry SAMs, so in scenarios V-11 and V-22 it says treat infantry platoons as if they have an infantry SAM squad. This rule helps with stacking (since attached squads do not count for stacking) and with keeping your precious FACs and SAM teams anonymous. It also works nicely with the transport rule, since attached squads do not need their own transport but can ride with the platoon.

97. **Damage to Attached Squads.** The roll of 4– seems about right. The probability of killing an attached squad is 40% for 1D and 64% for 2D. Those numbers are fairly good approximations to 1/3 and 2/3. If the roll was 3–, they would be 30% and 51%, and that seems a little low.

Are the damage rolls secret?

98. **Transportation.** The idea of transporting units and cargo comes from the "Thunder and Fire" scenario from *Eagles of the Gulf*. The rules here simply generalize it slightly, incorporate the idea of mobility, and add rules for mounting and dismounting.

99. **Mobility.** Being placed according to the mobility of its transport allows a tracked transport to place a towed unit in a forest hex.

100. **Mounting and Dismounting Infantry.** The rules for mounting and dismounting are completely new. I'm very open to suggestions about this.